

EXPERIMENTS IN ORGANIC CHEMISTRY

Third Edition



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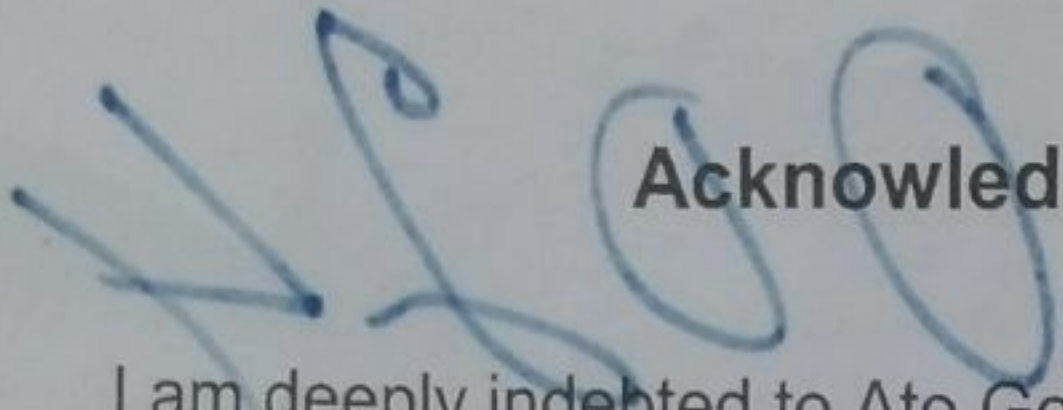
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By

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Addis Ababa, 2009



Acknowledgments

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FOREWORD TO THE STUDENT

This manual is intended for use in the first university laboratory course in organic chemistry. Such a practical course is usually offered simultaneously with an organic chemistry lecture course to beginning students of chemistry, medicine, biology, pharmacy, education, agriculture and chemical engineering.

Many of the techniques you learnt in general chemistry would be useful to what you will do and learn in this course. However, there are also several unique laboratory techniques that you will encounter in this course for the first time. You may also encounter some equipment and apparatus that are not used in other chemistry courses. You should therefore know the names of these apparatus and learn how to use and take good care of them.

It must be realized that this manual has to be supplemented by additional instructions, which will be relayed to you by your laboratory instructor. Depending on the situation and availability of chemicals, your instructor may modify some of the procedures given in this manual. If there is modification, this will usually be communicated to you at the beginning of a laboratory session. You should therefore take note of the changes and conduct the experiment accordingly.

In most instances you may be required to work in pairs, you must therefore make sure that you are an active participant in the group and not merely a bystander. Remember final assessments are based on individual performance. Clean all apparatus at the end of each experiment rather than before. Glassware is more easily cleaned after using it. Moreover, it is always better to begin an experiment with clean and dry apparatus. So when you are through with the experiment, please leave behind clean glassware.

Your final grade for this laboratory course will depend on how well you keep records during laboratory session, how well you write reports, your skills as will be monitored by continuous assessment by your instructor, quiz marks and result in the final examination.

Safety in the Laboratory

There are many potential sources of hazard in any laboratory. However this is more serious in situations where one works with chemicals and fire. Your instructor usually makes frequent announcements and warnings to inform you in advance about potentially dangerous situations. Please be attentive when such information is given and try your best to put the advice into practice. Most accidents can be avoided. Take care not only of yourself but also others who are working in the laboratory.

Our hope is that no accident happens in the laboratory. Take note of the following points.

Eye injuries

Strictly speaking one should never work in an organic chemistry laboratory without wearing safety glasses. Even visitors of a laboratory are expected to wear such glasses. Take all precautions so that no chemical gets into your eye. If this happens, report this immediately to your instructor and wash the affected eye(s) at once with a stream of water for as long as necessary. If you feel pain, make it a point to go to the clinic.

Fires and burns

Before using the Bunsen burner, learn how to handle it properly. Fire is one of the most common causes of accidents!

Do not light the burner by carrying lighted piece of paper from one part of the laboratory to the other, use safety matches or a gas lighter.

- Do not heat an inflammable liquid in an open flask directly with a flame.
- Learn how to use the fire extinguisher in your lab
- Small fire can be extinguished by using a piece of cloth or blanket
- Skin burns should be cooled with running water and not oil

How to get rid of waste

Do not dispose chemicals by pouring them into the sink or throwing them into waste baskets. Specially marked containers for this purpose are kept in the laboratory. Use them. Do not throw pieces of paper into sinks or on the floor. Use wastebaskets. Try to maintain as clean a desk as possible.

Cuts, burns and other injuries

Take note of the first aid kit and use them if need be. If reagents are spilled on the skin, flush immediately with water.

Your instructor will check if you are following these points at all time.

Advance Preparation

It is advisable to read the theory behind an experiment before coming to the laboratory. It is particularly useful to summarize the procedure in the form of a flow chart, which is best written using key phrases and arrows. Some instructors make it a point to check at the beginning of a laboratory session if students have come to the laboratory with a prepared flow chart. Some times a quiz is given at the beginning of the laboratory to check if the student has understood the theory.

About the notebook

Get yourself a bound note book. This notebook is the place where all data are entered. Do not use pieces of paper to record data as these could be easily lost or destroyed. Make a margin on each page and give page numbers. Get your note book checked before leaving the laboratory. Some instructors will insist that they sign on your note book as proof that you are present in the laboratory and also to follow up on what you have achieved in the lab.

About your report

You should submit a properly written report on the date

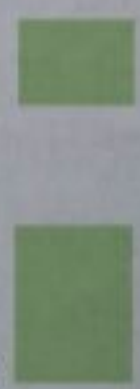
announced by your instructor. For this purpose use two to three sheets of A-4 size paper. The following summarizes the format for report writing:

- **Date:**
- **Experiment number:**
- **Title:**
- **Theory:**
- **Apparatus & materials:** List of apparatus and chemicals use in the experiment. In some cases a diagram of a set up may be useful.
- **Procedure:** Write the procedure in such a way that you or someone else can repeat the experiment by reading what is written here. Use the passive tense as described above.
- **Results and Discussion:** All data obtained are entered here. The Discussion Section is where you explain experimental observations. It is also important to give here the reasons behind some of the major steps followed while conducting the experiment. You must attempt, for instance, to give reasons why the yield of product you got is good or bad. You should also compare the actual yield you got with the calculated or theoretical yield.
- **Answers to Quiz Questions:**

Remember to write your report neatly, using good English and good handwriting.

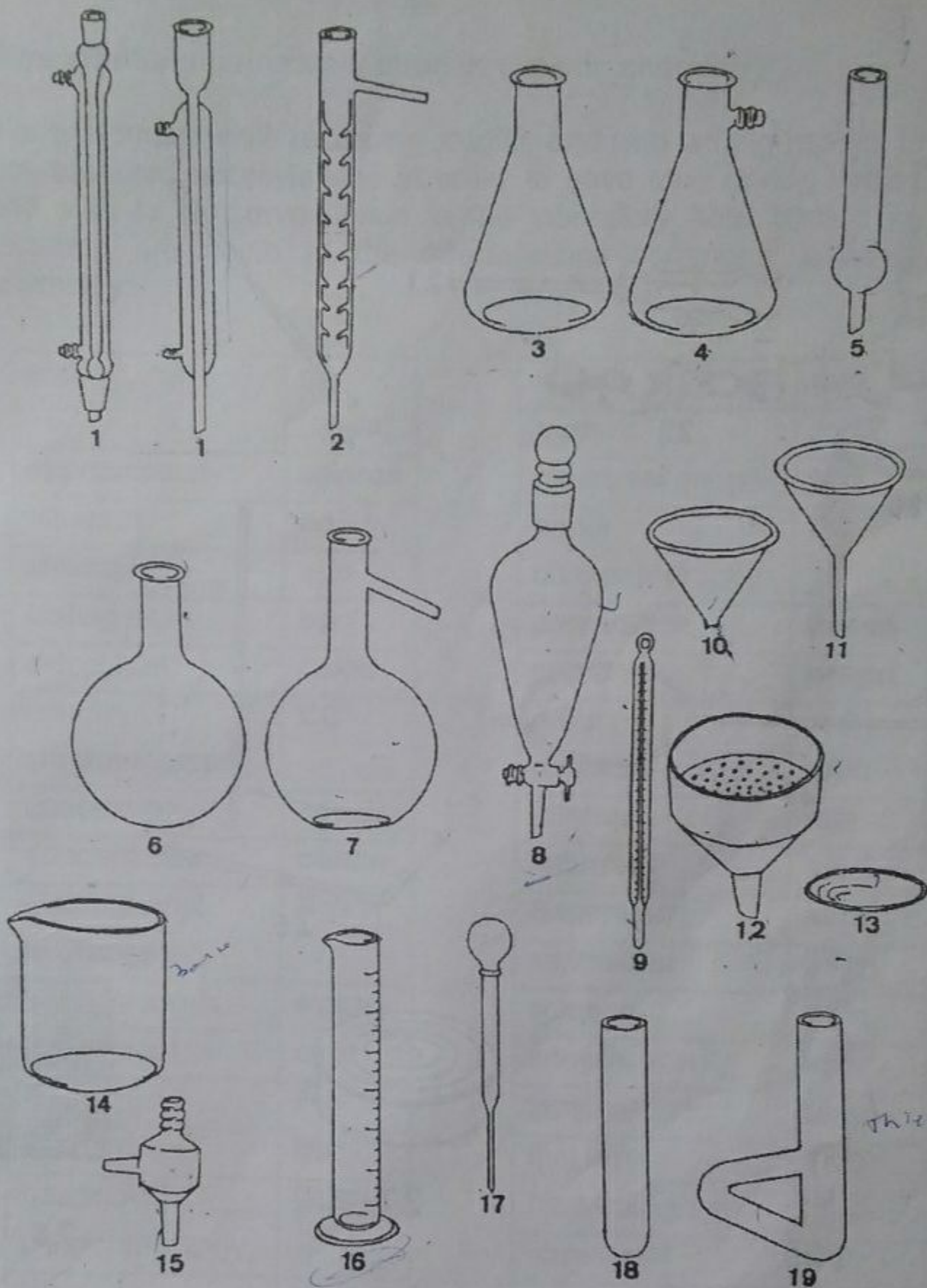
Signature

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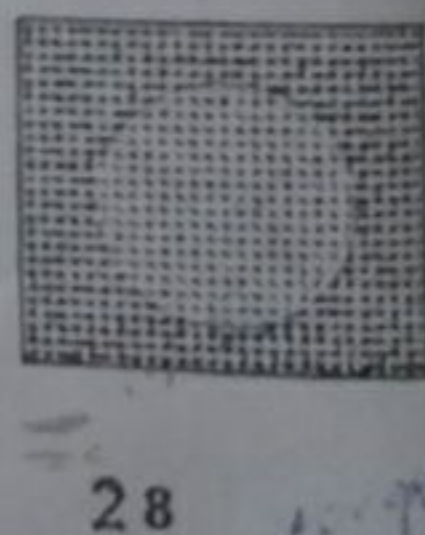
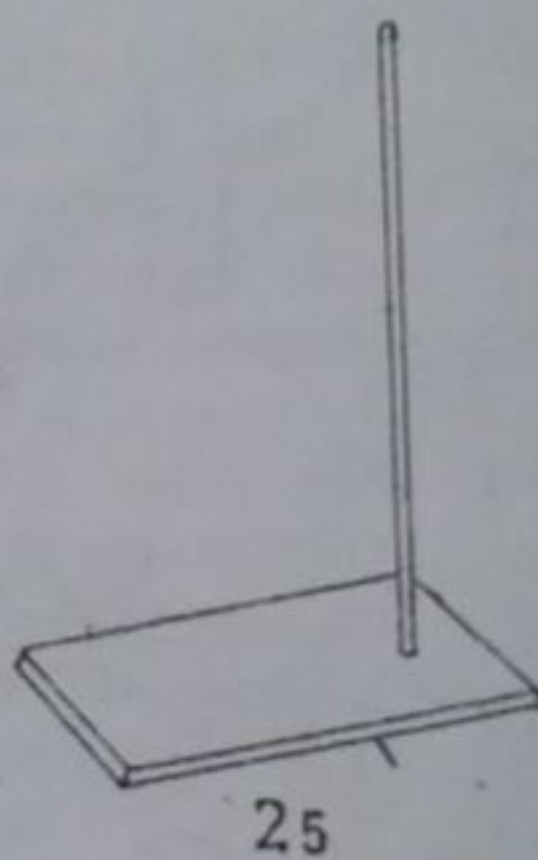
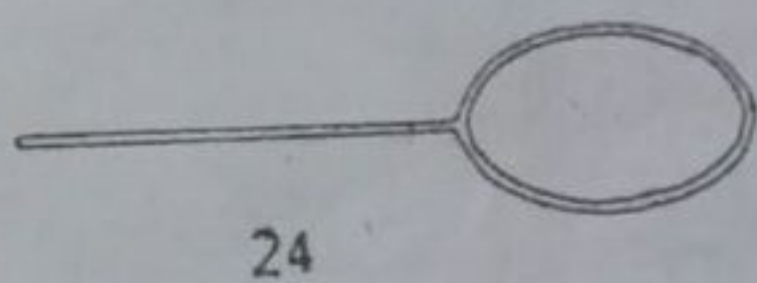
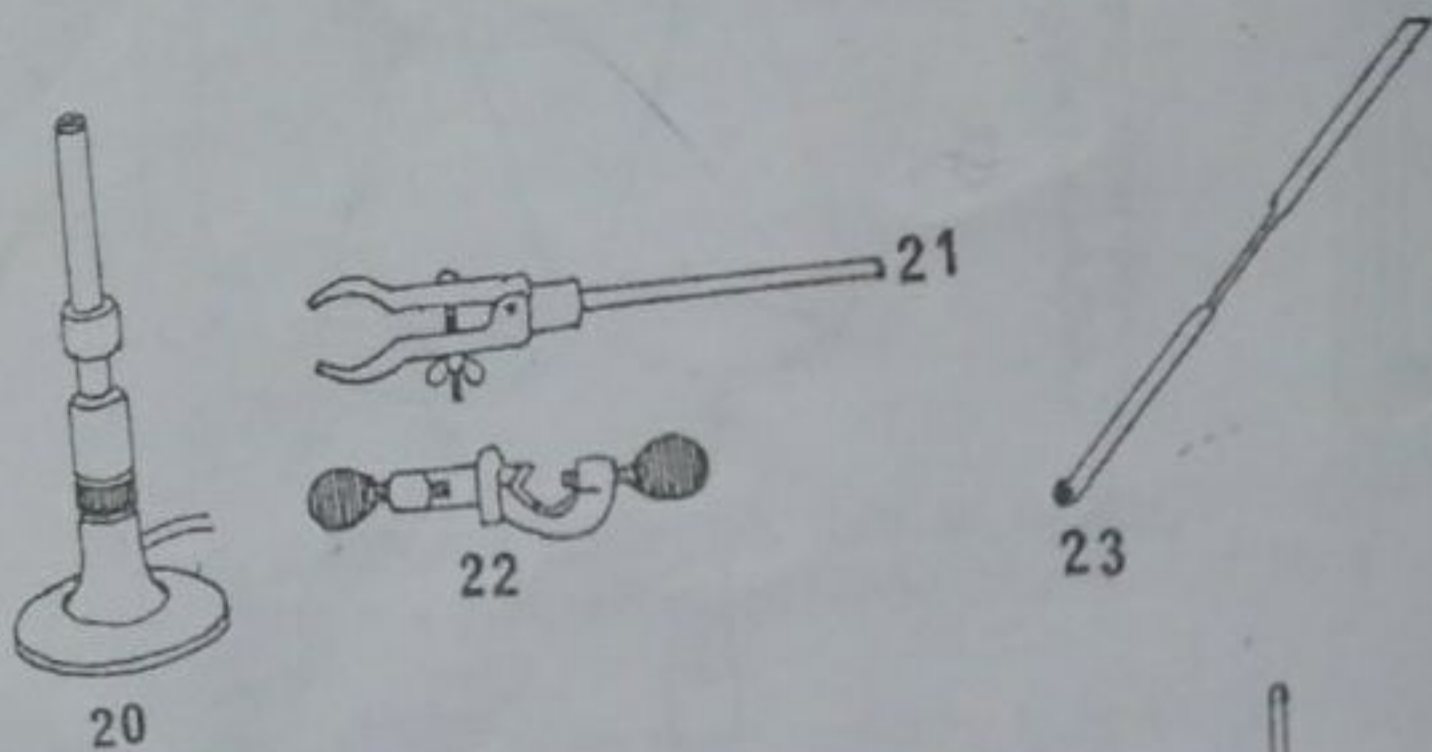
Always wear safety goggles for eye



1. Condensers
2. Fractionating column
3. Erlenmeyer flask
4. Suction or filter flask
5. Drying tube
6. Round bottom flask
7. Distillation flask

8. Separatory funnel
9. Thermometer
10. Stemless funnel
11. Long stemmed funnel
12. Buchner funnel
13. Watch glass

14. Beaker
15. Water aspirator
16. Measuring cylinder
17. Dropper
18. Test tube
19. Thiele tube



- 20. Bunsen burner
- 22. Bosshead
- 24. Iron ring
- 26. Test tube holder
- 28. Wire gauze

- 21. Clamp
- 23. Spatula
- 24. Iron stand
- 27. Water bath

Abbreviations commonly used in organic chemistry

It is customary while recording results, and also writing reports to use standard abbreviations in order to save time during writing and also to conserve space in the laboratory note book. The following are some of the abbreviations commonly used in chemistry:

about	ca (circa)
approximately	approx
aqueous	aq
atmosphere	atm
boiling point	bp
calculated	calcd
column chromatography	CC
concentrated	concd
concentration	concn
dinitrophenyl hydrazine	DNPH
evaporation	evapn
experiment	expt
figure	Fig.
fractions	frs
gas liquid chromatography	GLC
hour	h
infra red	IR
melting point	mp
milliliter	mL

minute	min
mixture	mixt
molecular weight	MW
paper chromatography	PC
preparation	prepn
prepared	prepd
reduced pressure	red pres
reference	ref
saturated	satd
Separated	sepd
separation	sepn
solution	soln
specific gravity	spgr
temperature	temp
thin layer chromatography	TLC
ultra violet	UV
volume	vol
weight	wt

Experiment 1

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Recrystallization

Objective

To purify a contaminated solid compound by recrystallization.

Theory

The purification of solids by crystallization is based upon differences in their solubility in a given solvent. In its simplest form, the crystallization process consists of:

- Dissolving the impure substance in some suitable solvent at or near the boiling point and filtering the solution while it is still hot,
- Allowing the hot solution to cool thus causing the dissolved substance to crystallize out and
- Separating the crystals from the solution.

Please note the importance of filtering the solution while it is still hot. This step is an additional purification process, since insoluble impurities will be separated by filtration. A most suitable solvent for recrystallization should:

- Dissolve the substance to be purified at elevated temperature but not at room temperature,
- Dissolve the impurities readily or to only a very small extent,
- Yield well-formed crystals of the purified compound,
- Not react with the substance that is to be recrystallized
- Be removed readily from the purified compound.

Some of the common solvents used for recrystallization include:

- For non-polar substances: Hexane, petroleum ether, chloroform

- For polar substances: Ethyl acetate, acetone, methanol, ethanol, water and acetic acid.

How to choose an appropriate solvent: The choice of solvent cannot always be made on the basis of theoretical considerations alone. It must be experimentally determined particularly if no information is available. About 0.1 g of the powdered substance is placed in a small test-tube and a solvent is added drop wise with continuous shaking. After about 1 mL of the solvent has been added, the mixture is heated to boiling, taking due precautions if the solvent is inflammable. If the sample dissolves easily in 1 mL of cold solvent or upon gentle warming, the solvent is unsuitable. If all the solid does not dissolve, more solvent is added in 0.5 mL portions, and again heated to boiling after each addition.

If 3 mL of solvent is added and the substance does not dissolve even upon heating, then another solvent should be sought. If on the other hand the substance dissolves upon heating, the tube should be cooled to see if crystals will form. To aid the crystallization process the inside of the test tube may be scratched with a glass rod. The fine scratches on the walls of the test tube usually serve as nuclei for crystal growth.

If a substance is found to be too soluble in one solvent and insoluble in another, a mixture of two or more solvents may be used for recrystallization. Pairs of liquids which could be used for this purpose include: alcohol and water, chloroform and petroleum ether, ethyl acetate and petroleum ether etc. When considering solubility of a substance in an organic solvent remember the principle "like dissolves like". In general non-polar substances readily dissolve well in non polar solvents, while polar substances readily dissolve in polar solvents. For substance with intermediate polarity combination of non-polar and polar substances is required. In this experiment you will purify a contaminated ("dirty") sample of an organic compound. The sample is made up of 90% pure compound with about 10% impurity. The main purpose of the experiment is to get rid of the impurities and recover the pure compound. Depending on availability of materials, you will be

given one of the following impure compounds: salicylic acid, aspirin, acetanilide, benzoic acid etc. Find out which the name and structure of the compound you are given to purify. All of the above substances can be recrystallized from water. So you do not have to attempt to find an appropriate solvent in these cases.

Procedure

In this experiment you will purify contaminated ("dirty") sample containing an organic compound and one or more impurities. Usually one of the following substances are the major present in a given sample: salicylic acid, aspirin, acetanilide, benzoic acid etc. These compounds are all readily recrystallizable by using water.

Your instructor will tell you what kind of substance you will use for this experiment. Please note the name of the substance in your note book.

Before starting the experiment, make a fluted filter paper. Your instructor will demonstrate how this is best done (See also Fig. 1). A fluted filter paper enables one to filter a solution at a faster rate than a simply folded filter paper.

➤ Weigh 1 g of the impure sample and place it in an Erlenmeyer flask. Add about 20 mL of water and heat the flask gently over a wire gauze with a Bunsen burner. Allow the water to boil gently. Place 50 mL of water in a measuring cylinder and from this add to the boiling mixture 5 mL at a time until almost all of the added sample has dissolved. Some insoluble impurities will remain suspended. In your note book keep a record of the total amount of water added up to this point. Stop heating the flask by turning off the gas connected to the Bunsen burner.

Place the fluted filter paper on a stemless funnel and put it on a flask. Now filter the hot solution by pouring it in small portions, keeping the apparatus as hot as possible. It may help to warm the funnel by placing it in the mouth of the hot flask or by heating the entire set-up. Any precipitate which forms may be dissolved again by adding small portions of boiling water. Do not add more water at this stage, because addition of more water will prevent the

formation of crystals. Cool the filtrate to room temperature. If crystals do not form at this stage, cool this further by placing it in an ice bath. Scratch the flask with a glass rod if necessary.

Take a new filter paper, **weigh** it (record!), and place it on the Buchner funnel, **wet it**, then attach the Buchner funnel to the suction flask, connect the water aspirator to it, turn the tap water and filter the crystals by pouring all of the contents of the beaker into the Buchner funnel. Your instructor will demonstrate how you to make the paper box. Make an open paper box, write your name and section on it and put the filter paper with your sample in the paper box till the next laboratory session. During this period your sample will dry. When you come to the laboratory **next week**, the first thing you will do is to weigh the dried filter paper + the sample (not the paper box). Subtract the weight of the filter paper in order to determine the weight of your product. You can then determine what the percentage of the impurity is in the sample that you recrystallized and you will then report it as **percent yield**.

Before leaving the laboratory:

- Clean all apparatus and put in their proper places
- Get signature of your instructor by showing the data that you have recorded

Before coming back to the next laboratory

- Write the report for the above experiment on 2-3 sheets of paper. Follow instructions in the Foreword section.

Here are some hints on how to write a good report: Your report should have the following sections: **Date, Title, Objective, Theory, Procedure, Results and Discussion, and answers to quiz questions.** You should write the theory briefly. For the **Procedure** part, write briefly what you did but using the passive tense instead of the active tense. For example "1 g of the impure solid was placed in the flask" but not "I took 1 g of the impure sample". Remember also to use the past tense. The most important part of the report is the **Results and Discussion** Section. Here you write the main results of the experiment and if your results are similar or different from what was expected. If there is some change you should also give some explanations.

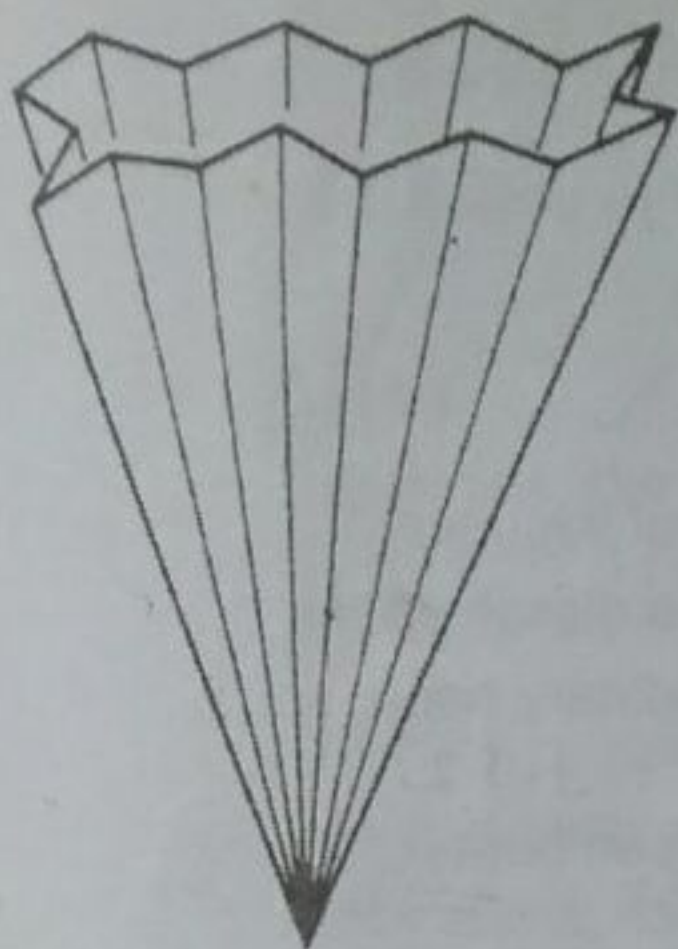


Fig. 1: Fluted paper

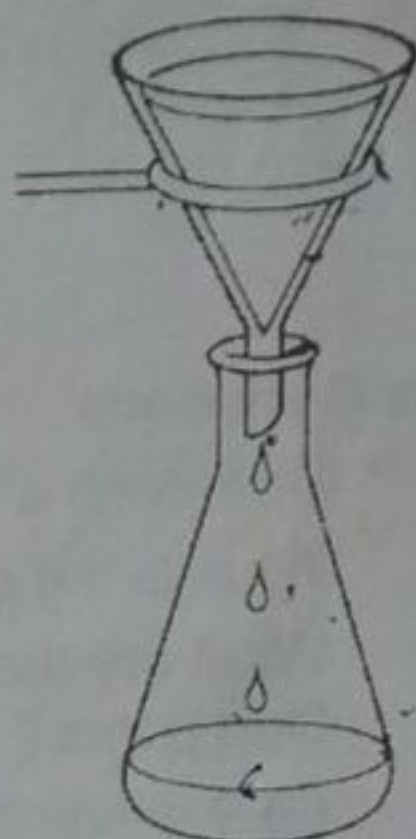


Fig.2: Gravity filtration set up

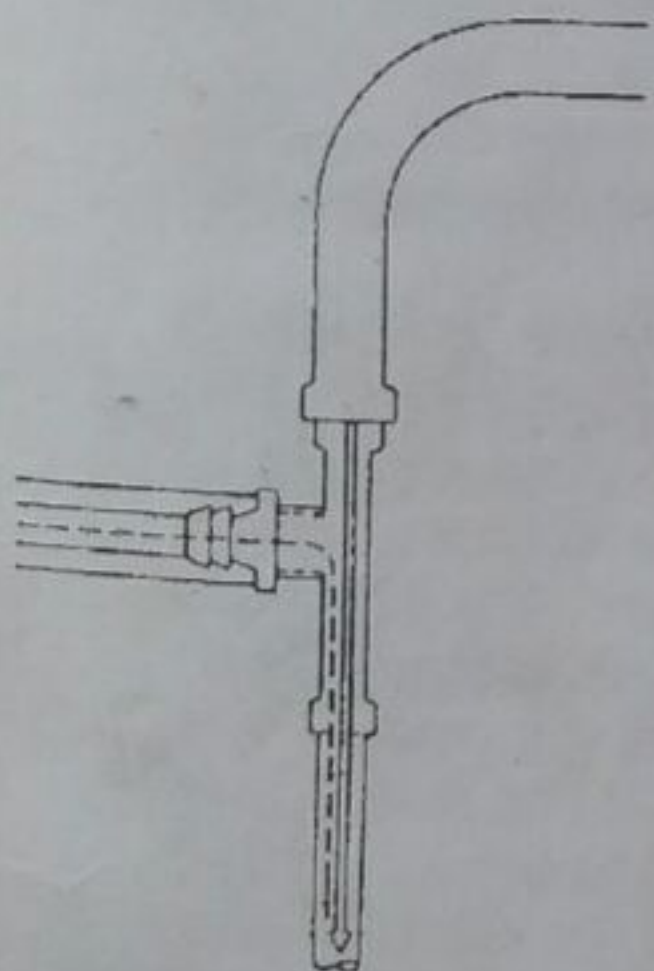


Fig. 3: Water aspirator

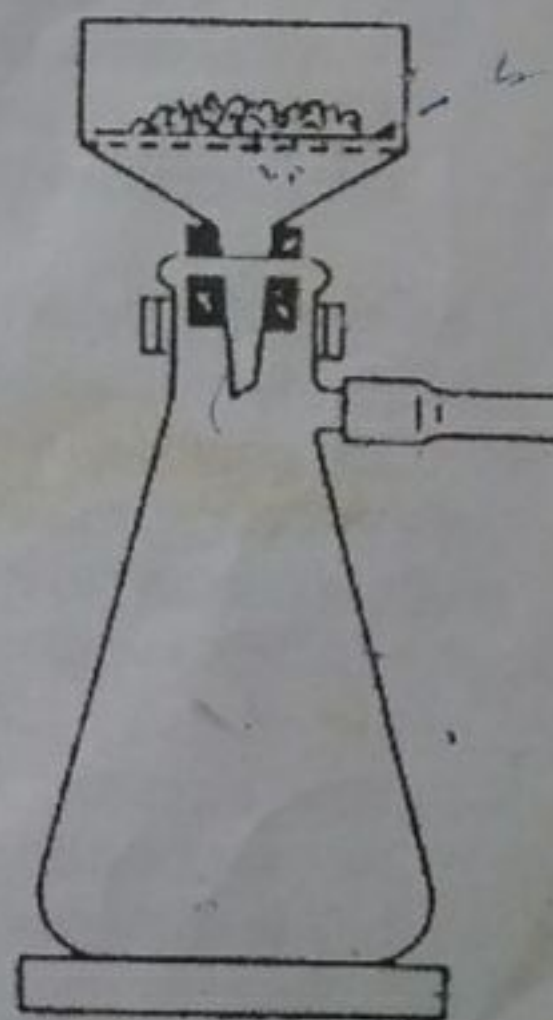


Fig. 4: Suction filtration set up

Threaded. 5
 Thread. 1000
 1000

Quiz Questions

1. How does a water aspirator create partial vacuum? ✓
2. What is the significance of the term "Like dissolves like"? ✓
3. What are the differences between an ordinary paper and filter paper? 0.82 1.52
4. Look at the flame generated by the Bunsen burner. What are the different parts of the flame? Which part is hottest? ✓
5. Some solids make perfect crystals, while others are amorphous. What may be the cause for this?

A & H

$$\begin{array}{r} 1.52 \\ 0.82 \\ \hline 0.7 \end{array}$$

Experiment 2

Determination of Melting Point (mp)

Objective

To determine the melting point (**mp**) of two samples: a pure compound and a recrystallized sample

Theory

The melting point of an organic solid is determined by introducing a small amount of the substance in a capillary tube, which is then attached to a thermometer partially immersed in an oil bath as shown in **Fig. 2.1**. The oil is heated slowly till the solid substance completely melts. One should carefully note the temperature at which melting starts and continue to observe in order to be able to note the temperature at which the melting process ends. Hence melting point is reported as a range e.g. $134-136^{\circ}\text{C}$. This indicates that the substance was melting between started to melt at 134°C and 136°C . A pure sample has a sharp mp with very small range while an impure substance has a lower mp with a broad range. A narrow range ($0.5 - 1.0^{\circ}\text{C}$) of mp indicates that a given sample is pure. On the other hand a melting range greater than 2°C indicates the presence of impurities. This is because contaminants depress the mp and broaden the melting range.

Furthermore, since the mp of a substance is characteristic of its nature, determination of its mp aids in its identification. Therefore mps are generally taken for two reasons, i.e. for checking the identity of a substance and for the determination of purity. In the latter case a technique known as determination of "mixed melting point" (mmp) is also used. In this method an unknown sample **A** is mixed thoroughly with another known sample **B**. If the mp of the mixture of **A** and **B** is the same as the mp of **A**, adding **B** does not depress the mp of **A** and therefore **A** and **B** must be identical. However if the mp is depressed then **A** and **B** are different.

Procedure

In this experiment you will first determine the mp of a **known pure compound**. You will be given **one of the following** substances: acetanilide (mp 113-114°C), benzoic acid (121-122°C), cinnamic acid (132-133°C) or salicylic acid (158-159°C). Then you will determine the mp of the substance, which you have purified by recrystallization in **Experiment 1**.

Apparatus: A very simple apparatus used to determine melting point is known as the **Thiele Tube** and is shown in Fig. 2.1. There are also very sophisticated equipment for the same purpose that are electronically controlled which give very accurate results.

Sample preparation: Make sure that the crystals are dry. Add a small amount of the sample into the provided **capillary** or **melting point tube**. This is done by placing a small amount of the sample on a piece of paper. Thrust the open end of the capillary tube into the sample so that some solid is trapped in the tube.

Turn the tube over i.e. closed end down and try to pack the solid down by tapping on the bench top or dropping the tube down a **long glass tube**. When the capillary bounces on the bench top the compound will be forced into the closed end. If you think there is not enough sample in the tube add some more. But remember that you should use the smallest amount of material that can be seen. Large samples usually heat unevenly. The capillary is held to the thermometer by a small rubber band made by cutting a thin slice off the end of a piece of tubing. The sample should be close to the end of the bulb of the thermometer.

Place the mp tube attached to the thermometer in the **Thiele tube**. Make sure that the **rubber band** is well **above** the surface of the **oil**. It should not come in contact with the oil, because it may spoil the oil. Get your set up **checked by the instructor** and then start to heat the arm of the Thiele tube gently. The hot oil will gradually circulate and will heat the sample in the capillary tube evenly. The hot oil goes up the side arm warming up the

thermometer and the tip of the capillary tube. The oil becomes cooler in the process and falls to the bottom where it would be warmed again, and this cycle continues as you do the experiment. If the approximate mp of the sample is known, you may first heat the oil at a faster rate, but when you reach about 20°C below the mp, the heating must be slow (2°C rise per minute). Remember that a faster rate will not give a good result and too slow a rate will cost you a lot of time. Watch changes of the solid sample in the capillary and also changes of temperature by following the thermometer reading. When the sample starts to fall apart and when liquid starts to form, record the temperature as the **start of melting** and when all the solid melts record that as the **end point of melting**. You can now report the mp as a range. Since you know the name and structure of the substance whose mp you determined, compare your result with the **reported mp** of the substance.

Caution: Do not get any water into the Thiele tube. Water should not be added to hot oil. If you have to cool a thermometer do not do so by dipping it in water simply let it stand in the open for some time. In the Thiele tube, make sure that the **rubber band** remains **above** the surface of the oil.

Melting Point Tube = Capillary Tube

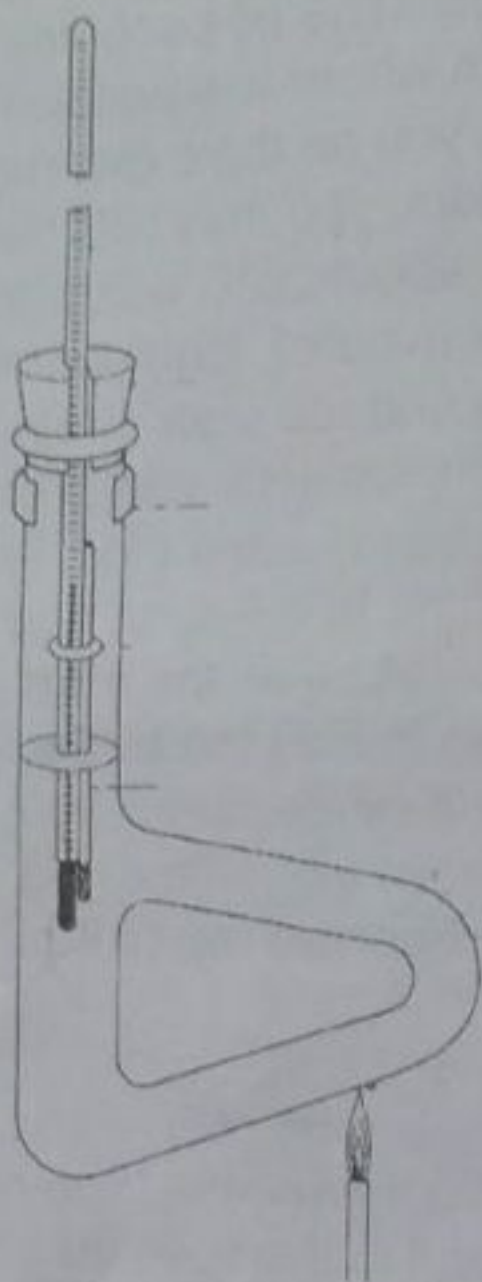


Fig. 5: Thiele MP apparatus

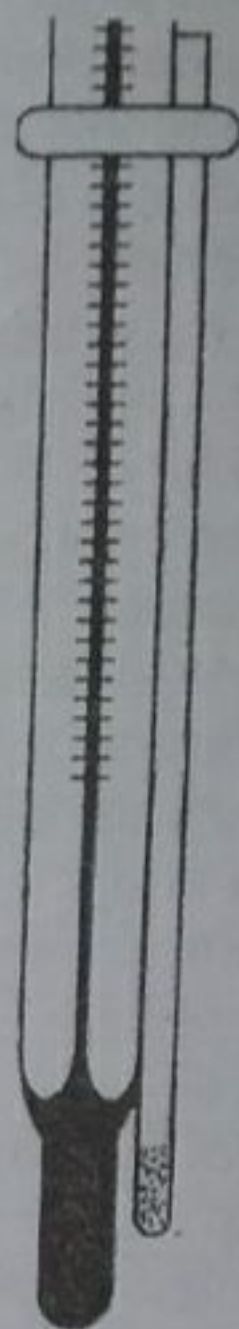


Fig. 6: Mp tube

Quiz Questions

1. Why is mp often reported as range and not as exact figure?
2. Why does an impurity reduce the mp of a substance?

Experiment 3

Simple Distillation

Objective:

To purify a contaminated liquid sample by simple distillation

Theory

Contaminant non-polar
Distillation is one of the most common methods of purifying liquids. If a liquid sample is contaminated with impurities that are non-volatile or if it is contaminated with another liquid which boils at least 50°C higher than the boiling point (**bp**) of the desired liquid, the main liquid can be obtained in pure form by simple distillation. Thus simple distillation is useful when only a small amount of impurity is present or when the liquids to be separated have widely differing boiling points.

Simple distillation is carried out by using a distilling flask to which a thermometer is fitted by means of a cork or a ground glass joint. The side arm of the flask is attached to a condenser. As the flask is heated vapours rise into the neck of the flask and at first it may condense and run back and then gradually the vapour passes into the condenser where it is reconverted to liquid. As the distillation proceeds the bulb of the thermometer carries drops of the condensate and the temperature reading on the thermometer is recorded as the **boiling point of the liquid**. An ascending stream of cold water in the jacket of the condenser facilitates the condensation process. The condensate is then collected in a receiving flask. The non-volatile higher boiling fraction is left as a residue in the distilling flask. Distillation can also be done by using a vacuum pump which reduces the boiling point of the liquid to a great extent.

Some liquids tend to superheat, that is they rise to a temperature slightly above their boiling point. This is a meta-stable condition which might lead to violent eruption or bumping caused by the sudden escaping of vapour out of the super heated liquid. This phenomenon may complicate the distillation process and should

therefore be avoided as far as possible. **Bumping** is avoided by adding **two or three boiling stones or boiling chips before starting the distillation**. In the presence of these small pieces of stone or fine silica chips, boiling proceeds smoothly. One should therefore not boil any liquid without adding boiling chips.

Procedure

Set the distillation apparatus by looking at the set up prepared for demonstration. Pour 25 mL of the provided ethyl alcohol which has intentionally been contaminated with a dye and higher boiling liquid (e.g. glycerin or ethylene glycol etc). Use a funnel when adding the alcohol into the distilling flask. Add 2 or 3 pieces of **boiling chips** (Why?). The distillation is performed **not** by directly heating the flask **but** by placing it **on a water bath** which will be heated by means of a Bunsen Burner. The condenser should be attached to the arm of the distillation flask. The condenser has an inlet and outlet for water. Make sure that the **lower** is the **inlet** that should be attached to the water tap using a rubber or plastic tube. Place the thermometer so that its bulb is about 0.5 cm below the junction of the side arm of the distilling flask.

Turn the water tap gently. Place a 25 mL measuring cylinder for collecting the distillate. Get your setup checked by the instructor. Light the Bunsen Burner and observe the distillation process carefully as described in the Theory Section above. During the distillation the thermometer bulb bears drops of condensate. Record in a Table the boiling point after **every 2 mL**. When most of the volatile alcohol has been distilled off, the rate of drops of liquid coming into the receiving measuring cylinder decreases. After some time no more drops will come out. This indicates that you should stop the distillation by turning off the Bunsen Burner. Record the total volume of alcohol that you have collected and calculate the percentage yield of the distilled alcohol. Plot the results using a graph paper volume (x-axis) vs. temperature (y-axis) and discuss the results in your notebook.

Before leaving the laboratory:

- Take apart the distillation setup
- Handover the distillate and the residue to your instructor
- Clean the flask and put the apparatus in their proper places
- Get signature of your instructor by showing the data that you have recorded

Before coming back to the next laboratory

- Write the report for the above experiment on 2-3 pages of loose leaves of paper.

Here are some hints on how to write a report: Your report should have the following sections: **Date, Title, Objective, Theory, Procedure, Results and Discussion, and answers to quiz questions.** You should not write much for the Theory part, simply say **see the manual**. For the **Procedure** part write briefly what you did but using the passive tense instead of the active tense. For example "25 mL of impure liquid was poured into the flask" but not "I poured 25 mL of impure liquid". The most important part of the report is the **Results and Discussion** Section. Here you write the main results of the experiment and if your results are similar or different from what was expected. If there is some change you should also give some explanations.

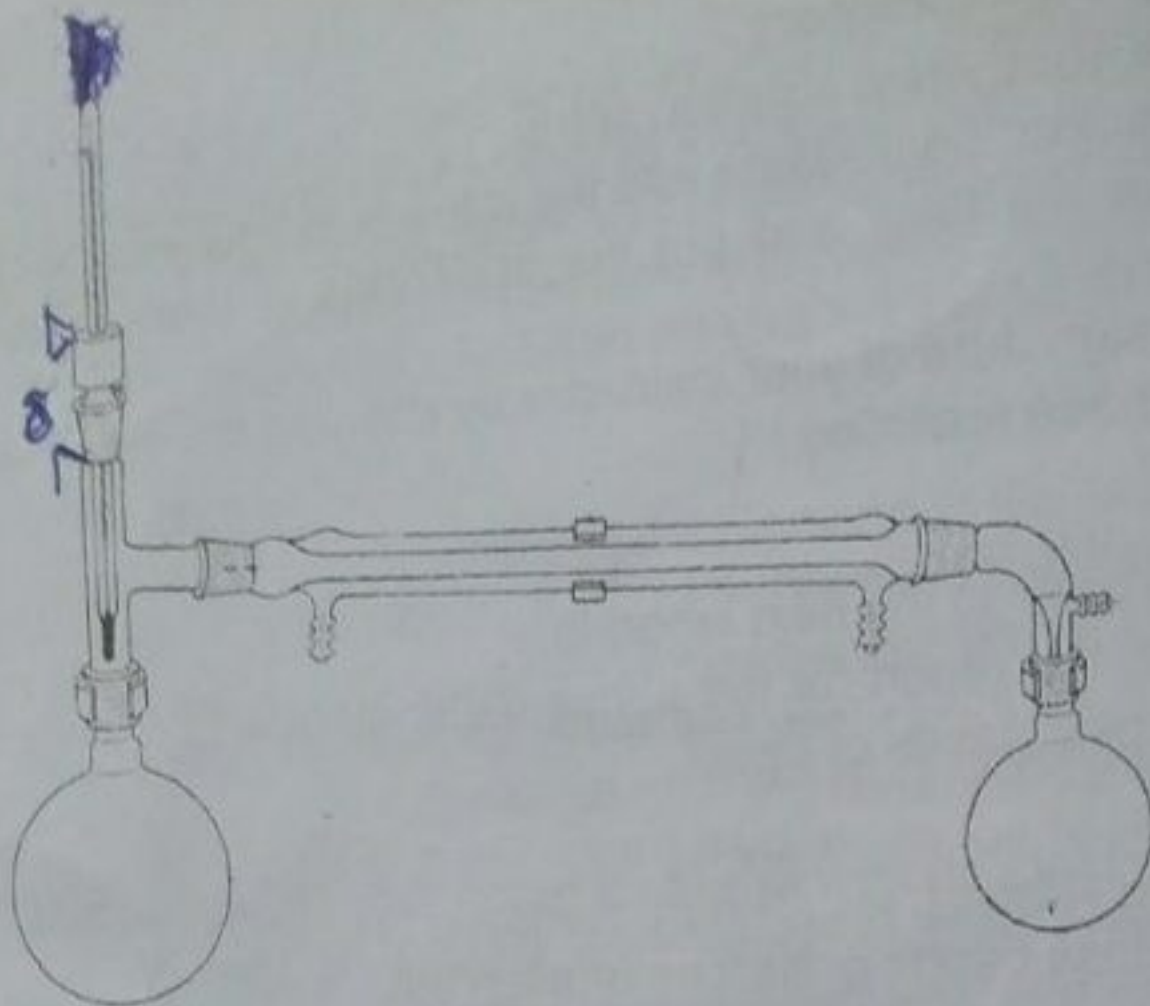


Fig. 7: A typical set up for simple distillation

Quiz Questions

1. Why does not all the liquid vaporize at once when the boiling point is reached?
2. An organic liquid begins to decompose at 80°C . Its vapour pressure at that temperature is 36 mm. How would you distill the liquid?
3. Why is cold water circulated through a condenser from the bottom upward rather than from the top downward?
4. If the liquid to be distilled had a boiling point higher than 90°C , what kind of bath would you use?
5. Why is it that when a pure liquid is boiled its temperature remains constant?

Experiment 4

Fractional Distillation

Objective

To separate a mixture of two liquids by fractional distillation.

Theory

Mixtures of volatile liquids can be separated into their component parts by a technique known as fractional distillation. In this process volatile liquids, which boil within 25 °C of each other are separated into components, which are called fractions. The separation is made possible by a fractionating column, which is attached to a distilling flask. The fractionating column may be packed with glass beads or helices or even other solid pieces like ceramic, metal chips etc. The presence of these materials in the column causes several tiny distillations to occur on the surface of the column packing.

In order to understand the principles underlying fractional distillation, we will briefly review the relationship between vapour pressure and boiling point.

If the solution of two miscible components A and B obeys Raoult's Law, the contribution that each component makes at a given temperature to the total vapour pressure can be calculated:

$$P_A = X_A P_A^0 \quad P_B = X_B P_B^0$$

Where X indicates mole fraction.

$$P_{\text{solution}} = P_A + P_B$$

When a mixture of two liquids is boiled the vapour normally contains a higher proportion of the more volatile (i.e. lower boiling component) and the residue contains a higher proportion of the less volatile component. The amount of each substance in the vapour depends on the contribution each liquid is making to the total

vapour pressure at the boiling point of the mixture. The vapour moves upwards, comes in contact with the packing material in the fractionating column and partially condenses. The condensate is richer in the less volatile component, as the vapour that continues to rise is richer in the more volatile liquid. A number of such equilibrations take place in the fractionating column, resulting in the rise of the vapour of the more volatile component while the liquid of the less volatile component runs down the column.

The composition of the liquid remaining in the flask changes gradually by becoming richer in the higher boiling component. The boiling point of the mixture therefore rises as the distillation proceeds. If the distillate is collected in fractions, and the fractions redistilled almost complete separation is achieved. Separation is, of course, easiest if there is a large difference in boiling point between the two liquids. This is the essence of fractional distillation because the process of repeated distillation is carried out in a simple way with the help of the **fractionating column**, which in effect causes several tiny distillations.

Procedure

Observe the setup for this experiment as arranged by your instructor. This experiment requires that the fractional distillation will be performed by placing the flask on water bath. This set up differs from that for simple distillation by the introduction of a fractionating column.

Pour the provided 50 mL of ethanol-water (1:1) mixture into the distilling flask. Introduce two or three boiling chips, and get the set up checked by the instructor before starting the fractional distillation. Collect the distillate directly into a measuring cylinder and record the temperature after every 2 mL. When the measuring cylinder reading reaches 12 mL (**Fraction 1**), change the receiver to an Erlenmeyer flask and continue to distill till you receive another 12 mL (**Fraction 2**). The temperature will then begin to fall down. You should then remove the water bath, in its place put a wire gauze on the iron ring and heat the flask through the wire gauze with a gentle flame. Change the receiver and

Quiz Questions

1. What is an azeotrope? (Describe this phenomenon by using the distillation of mixture of benzene, ethanol and water, in the preparation of absolute ethnaol.)
2. If liquids A and B both have a bp of 160°C and do not form an azeotrope what is the bp of a mixture of the two liquids? What does this indicate about using bp as an indicator of purity?
3. Why does the temperature fall down towards the end of the fractional distillation?
4. What is the use of a fractionating column and how is it made?

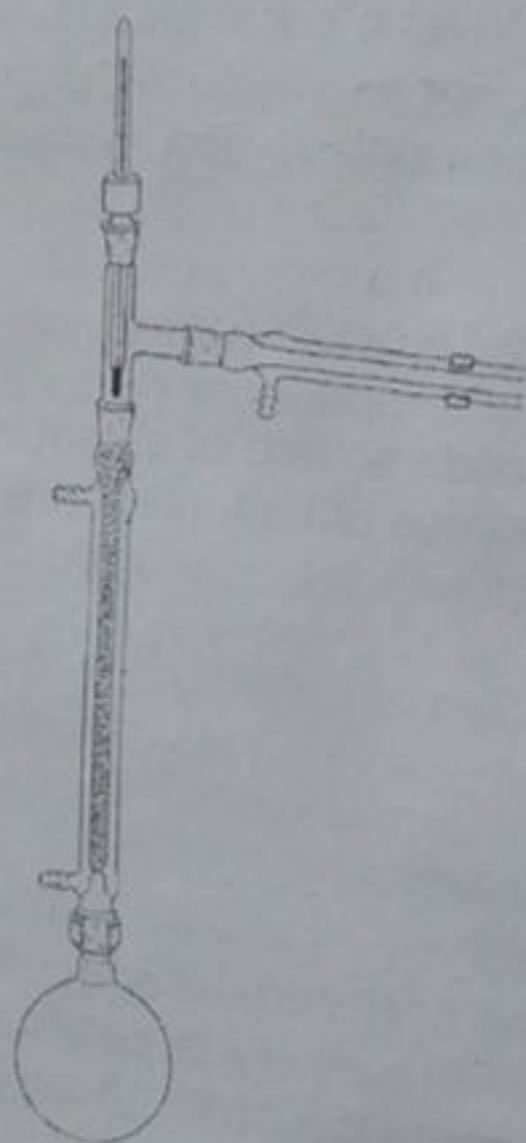


Fig. 8: Set up for fractional distillation

Experiment 5

Steam Distillation (Hydro-distillation)

Objective

To separate slightly volatile but water insoluble substances from non-volatile materials.

Theory

In experiments 3 and 4 two or more miscible liquids were separated into their component parts by using simple and fractional distillations. We will now consider the distillation behavior of two-phase systems consisting of two immiscible liquids. Steam distillation is particularly useful for separating a product, which is insoluble in water from a complex mixture of non-volatile substances such as organic salts, unreacted starting materials or side products etc. For example, in the preparation of aniline by the reduction of nitro-benzene with tin and hydrochloric acid, the reaction vessel finally contains a mixture of tin salts, alkali, water and aniline. The aniline is recovered in a fairly pure state from this mixture by means of steam distillation.

Steam distillation uses an outside source of steam which pipes the steam into the distillation unit, sometimes at high pressure. The steam passes through the material, and exits into the condenser.

What is the advantage of **steam distillation** when compared to **simple distillation**? Some high boiling substances decompose at their boiling points if they are subjected to simple distillation at atmospheric pressure. For instance the bp of aniline is 184°C . One way of purifying aniline is by distilling it, which requires heating it to its bp. If this method is followed some of the aniline may decompose at the high boiling temperature. However by using steam distillation aniline can be easily separated from a mixture at the boiling point of water (100°C) which is much lower than the boiling point of aniline.

The answer is to be found in the fact that in a system of two immiscible liquids, say A and B, the vapour pressure of each at a particular temperature is the same as that of the pure liquid alone. The mixture boils when the sum of the two vapour pressures equals the atmospheric

record the temperature after approximately every 2 mL, and collect 12 more mL of distillate (**Fraction 3**).

Optional: If a digital balance is available determine weight of empty 10 mL measuring cylinder. Then add 10 mL of Fraction 1 to it and weigh again. The difference will give you the exact weight of the liquid inside the measuring cylinder. You can now calculate the specific gravity of Fraction 1. Repeat for Fractions 2 and 3. What do the results indicate? Note that the sp gr of pure ethanol is 0.8.

Which of the distillates is richest in ethanol, and which one in water? Test for water in each fraction by anhydrous **copper sulfate test**. This is done by adding to 1 mL of each fraction placed in small test tube and adding to each small crystal of copper sulfate. Blue colour formation indicates presence of water.

Hand over the three distillates separately to your instructor and report the volume of each and the boiling point ranges during collection.

Tabulate your data and plot a graph (use graph or squared paper) showing the relationship of temperature (y- axis) and volume (x- axis).

pre-ether
pressure. This is expressed by the following equation:

$$P_A + P_B = P_{\text{atm.}}$$

The mixture has therefore a constant boiling point, which is independent of the relative amounts of A and B present and is also lower than that of either pure A or pure B.

Assuming ideal behavior, $P(A)$ and $P(B)$ are proportional to the number of molecules of A and B respectively present in the vapour phase:

$$P_A / P_B = n_A / n_B$$

Where n_A / n_B is the molar ratio of A to B in the vapour phase.

The ratio of weights of A and B in the vapour phase will depend not only upon the ratio of moles but also upon the molecular weights of A and B.

$$W_A / W_B = n_A \times MW_A / n_B \times MW_B = P_A \times MW_A / P_B \times MW_B$$

$$\frac{W(A)}{W(B)} = \frac{n(A) \times M.W.(A)}{n(B) \times M.W.(B)} = \frac{P(A) \times M.W.(A)}{P(B) \times M.W.(B)}$$

If the above type of distillation is conducted with water, **A** is water and **B** would be an organic compound. Since the molecular weight of water is low, a fairly high proportion of **B** is found in the distillate even if **B** is much less volatile than water. Thus the relative amount of water and the organic substance that are collected in the receiver during steam distillation depends on the vapour pressure of water and that of the organic substance at the distillation temperature and their respective molecular weights. Steam distillation leads to good separation if the insoluble substance to be steam-distilled has a vapour pressure of at least 5-10 mm at 100°C.

Another alternative method is **hydrodistillation**. By this method there is no outside source of steam. The material to be purified is mixed with water and the water is heated to boiling. The steam that is generated comes in contact with the volatile substances and the water vapour carries the volatile components with it. When the vapours are cooled by means of a condenser, the volatile components also condense with the water.

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and both are collected in a receiver. Since these components are not soluble in water, they separate from the water as a phase, which may be above or below the water depending on density. If one wishes to separate the organic layer from the water, it is necessary to use at this stage a separatory funnel. Hydrodistillation is a more ancient method of distillation and the most versatile. Hydrodistillation seems to work best for powders such as spices, ground wood, roots, nuts etc.

Apparatus

Your instructor will tell you which of the following two set ups you will use:

Hydrodistillation Assembly: As described above, the organic material is mixed with water in a distillation flask connected to a condenser and the mixture is made to boil. The boiling water will serve as internal source of steam.

Steam Distillation Assembly: A steam distillation set up requires generation of steam in a flask and leading the steam into a second flask containing the material to be steam distilled.

PROCEDURE

Experiment A: Place 5 mL of toluene, 100 mL of water and two boiling chips in a distilling flask. Heat the flask with gentle flame through a wire gauze and bring the water to boiling. Add more hot water periodically if needed. Collect the distillate in a measuring cylinder and stop the distillation when the distillate that comes over is no longer cloudy. This is best checked by collecting a few drops of distillate in a small test tube. This is because when an organic compound that is not soluble in water is mixed with water, it gives rise to milky or cloudy colour. Therefore when there is **no** more toluene coming out of the distilling flask the distillate water becomes clear, and the heating of the flask has to be stopped.

You will see two layers in the measuring cylinder i.e. the aqueous and the organic phase. Decide which of the two layers is the organic phase before proceeding to separate them. Salting out the organic compound, by adding small amount of common salt (NaCl) to the aqueous layer, and shaking may some times help to clearly separate the organic from the aqueous layer. Note down the volume of the organic layer using the reading of the measuring cylinder. Also note the volume of water that

came out during the distillation. Separate the two layers by using a **separatory funnel**. Place the organic layer in a flask and give it to the instructor.

When you write your report calculate the weight ratio of the organic layer, to that of the distillate water. To convert the volume of the organic compound to its corresponding weight, you need to know its density. If the organic compound used is toluene, note that its density is 0.86 g/mL. Compare the experimental ratio you obtained with the theoretically calculated value.

Experiment B: Essential oils are produced by either steam or hydro-distillation. In the case of steam distillation water is heated in a boiler to produce steam. The steam is led to a container that contains the material to be steam distilled. The steam then carries volatile components with it. The steam is then chilled in a condenser and the resulting distillate which contains lots of water and some essential oil is collected in a receiver. The essential oil will normally float on top of the distillate water and it can be separated from the water by using a separatory funnel.

Carefully observe the demonstration and note down the name of the plant, whether the method used is steam distillation or hydrodistillation, whether the distillation process is fast or slow, whether the essential oil is lighter or heavier than water etc.

Note to the Instructor: Use a Clevenger apparatus

Quiz Questions

1. What is the difference between hydrodistillation and steam distillation?
2. Why does boiling point of a liquid decrease with increase in altitude?
3. How does salting out work?
4. What are some of the uses of essential oils?

5. Experiment 6

Survey of Some Functional Groups

Objective: To study the characteristic chemical properties of some functional groups

Theory

A functional group is a unique combination of atoms present in a molecule that is particularly responsible for the characteristic set of reactions that the molecule exhibits. Some of the most important functional groups include: hydroxyl, olefin, ketone, aldehyde, amine, carboxylic acid, ester, acid halide, anhydride, nitrile, ether, halide, aromatic ring etc. It is possible to distinguish by means of simple tests one functional group from the other.

Alkanes, the simplest of all organic compounds, lack the typical functional groups making them most unreactive. They can be inert even in the presence of conc sulfuric acid. Alkanes react slowly or not at all with bromine in the dark. However in the presence of light react fairly rapidly. This can be explained by means of radical chemistry.

The simplest of alkanes is methane which is a gas at room temperature, so are ethane, propane and butane. Pentane and hexane are low boiling liquids and are among the most useful solvents in laboratory as well as in industry. Mixtures of alkanes obtained during fractional distillation of petroleum are called petroleum ether or simply petrol. They are usually categorized by means of their boiling point range. A low boiling petroleum ether has boiling range of 40-60°C, while the higher boiling ones are 60-80°C and 80-120°C.

Gasoline also known as benzine is one of the most common sources of energy and so are the higher boiling naphta or kerosene. These fuels are mixtures of alkanes such as isomeric pentanes, hexanes, heptanes and octanes and small amounts of unsaturated and aromatic hydrocarbons.

In this experiment attempts will be made to highlight some of the most important tests in organic chemistry that aid one to tell functional groups apart.

Procedure

A. Alkanes

Carry out the following tests in **dry** test tubes.

Solubility of Alkanes: Test the solubility of Gasoline in water, ethanol, petroleum ether (or hexane) and concentrated sulfuric acid. In each case place 5 drops of gasoline in a small, dry test tube and add not more than 3 ml of solvent. Observe and record your results in a table.

Repeat the above experiment for kerosene.

Reaction of Alkanes:

In three separate test tubes place 1 ml portions of petroleum ether, gasoline and kerosene and add to each 4-5 drops of a solution of bromine in carbon tetrachloride (5%). Place the three test tubes in the dark (i.e. in your cupboard). Observe the colour of each test mixture after 15 minutes and note your observation. The presence of HBr can be confirmed by holding a wet blue litmus paper on the mouth of the tube.

Repeat the above experiment, but this time take the three test tubes outside in the sun for 5 minutes or so. Compare the result with the previous experiment. Is there any difference? Explain the results. (Hint: Gasoline contains some unsaturated hydrocarbons.)

Observe the effect of oxidizing agents on the above hydrocarbons as follows: Place 10 drops of each of the above mentioned three hydrocarbons in three test tubes and add to each 1 ml of dilute (0.5%) potassium permanganate solution. Shake well, observe the result and record in a table using terms such as soluble or miscible, insoluble or immiscible.

After the experiments put the reaction solutions into the provided special container. Do not pour chemicals in the sink.

B. Alkenes

Solubility of alkenes: Place 0.5 ml (10 drops) of cyclohexene in a dry test tube. Check the solubility of this compound in concentrated sulfuric acid (3 drops).

Reactions of alkenes With bromine in carbon tetrachloride: To 0.5 ml (10 drops) of the given alkene add 3 drops of bromine/carbon tetrachloride solution.

With aqueous permanganate solution: To 0.5 ml of the given alkene add 1 ml of 0.5% aq. potassium permanganate solution. Shake well and record your observation. Explain all of the above results by means of chemical equations.

C. Alkynes

The simplest member of this class is acetylene and is widely used industrially. Acetylene is readily prepared according to the following reaction.



1. **Generation of Acetylene:** Carefully examine the acetylene generator which has been set up by your instructor in the fume hood. Draw a diagram of the generator in your notebook. Allow water drop by drop into the calcium carbide so that acetylene is generated gently.
2. **Bromination Test:** Bubble acetylene into a test tube that contains 1 ml of a bromine/carbon tetrachloride solution (0.25%). Observe and record the result.
3. **Baeyer's Test:** Bubble acetylene into a test tube containing 2 ml of dilute (0.5%) potassium permanganate solution. Observe and record the result.

D. Aromatic hydrocarbons

1. **Test for unsaturation:** To 6 drops of benzene or toluene add 3 drops of bromine/carbon tetrachloride solution. Observe if there is any reaction. If there is no reaction explain why aromatic compounds do not undergo the typical reactions of olefins.

2. **Nitration of benzene or toluene:** In a test tube prepare a nitrating mixture by adding 10 drops of concentrated sulfuric acid and 3 drops concentrated nitric acid. Add to these 4 drops of toluene. Shake the test tube and let it stand for 15 minutes and then pour the mixture into a large test tube containing a piece of ice. Nitrobenzene is formed as slight yellow oil. Nitrobenzene is toxic avoid contact. Give the equations with mechanisms.

E. Test for ketones

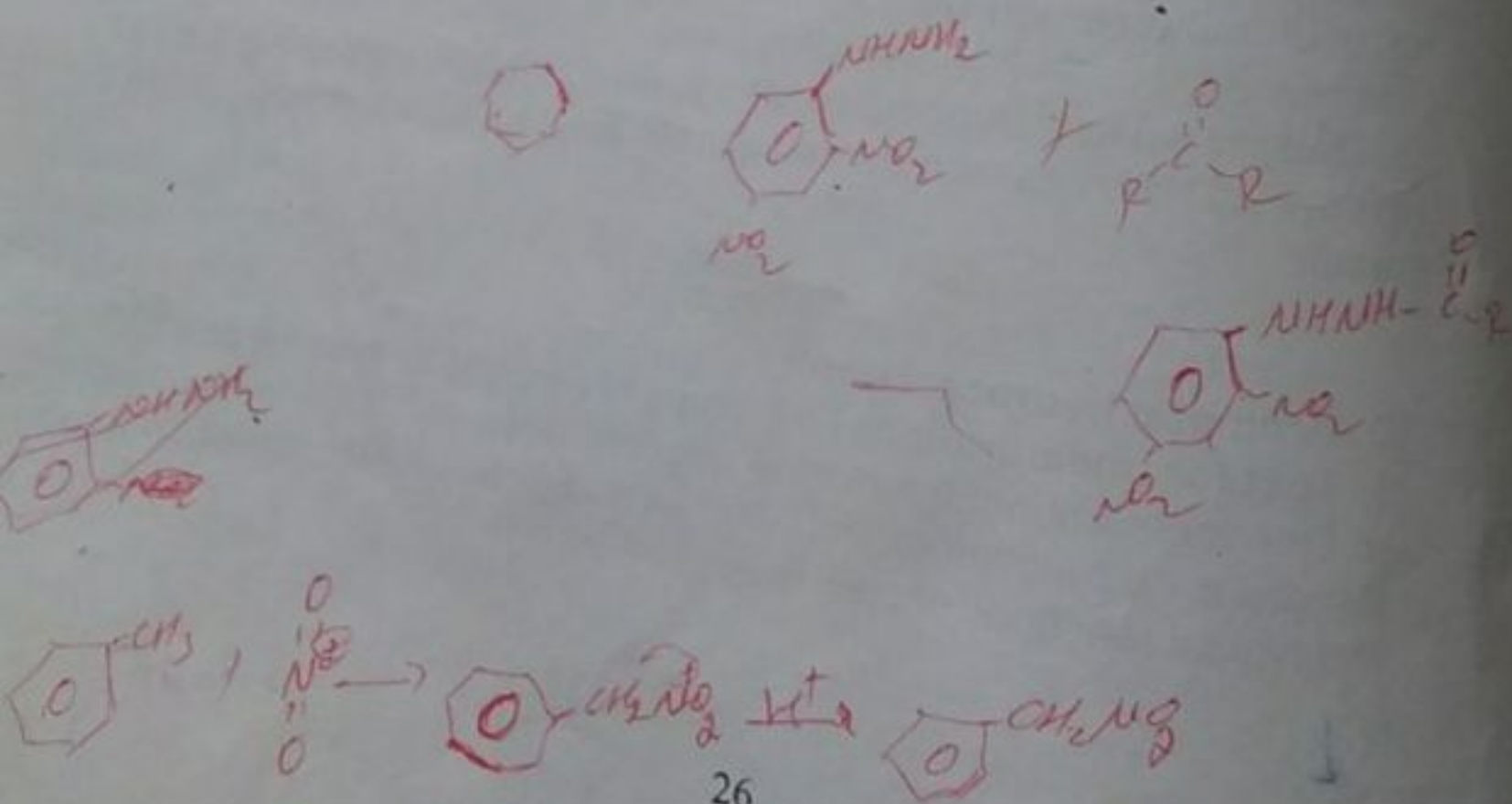
A typical reaction of ketones is their reaction with 2,4-dinitrophenylhydrazine (DNPH). Place 10 drops of the acetone solution in a small test tube. Add few drops of the reagent to it till you notice formation of precipitate. Write the chemical equation for the reaction.

F. Test for phenols

Place 20 drops of 10% aq solution of phenol in a test tube. Add to these 3 drops of 2% of neutral ferric chloride solution. The development of a violet color is characteristic of the phenol functional group.

Quiz Questions

1. Give the mechanism of the reaction of DNPH with acetone?
2. What is the difference between addition and substitution reactions?
3. Give mechanism for the nitration of toluene?



Experiment 7

The Third Dimension in Organic Chemistry: Stereochemistry

Objective: To observe three-dimensional structures of molecules

Theory

Review principles of stereochemistry by referring to your text book. And familiarize yourself with the following terms and concepts:

Achiral molecule	Dissymmetry	Seric hinderance
Asymmetric carbon	Eclipsed conformation	Structural isomers
Baeyer strain	Enantiomer	Symmetry element
Bond angle	Gauche interaction	Ball and stick model
Chiral compound	Geometric isomerism	Fischer Model
Configuration	Newman projection	Space filling mo
Conformation	Optical activity	
Conformer	Polarimeter	
Diastereoisomer	Racemic mixture	
Dihedral angle	Steric interaction	

Make sure that you understand the underlying principles that help one understand the difference between geometric isomerism and optical isomerism. Geometric isomerism arises when either a double bond or a ring is present in a molecule.

Two objects are said to be identical to each other if one can be superimposed on the other. A dissymmetric object cannot be superimposed on an object that appears as though it were its mirror image. Allow your right hand to face the left. They appear to be mirror images of each other. But it is not possible to superimpose your left hand on your right hand. This is because these two objects are dissymmetric. This concept also holds for molecules. If two molecules are dissymmetric and at the same time appear to be mirror images of each other, if you build models you can show that it will not be possible to superimpose one on the other. This phenomenon leads to optical activity.

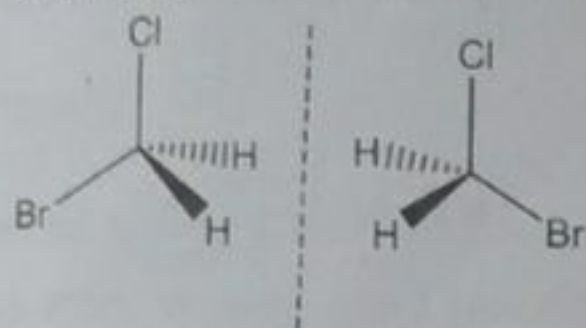
Procedure

1. Geometric isomers

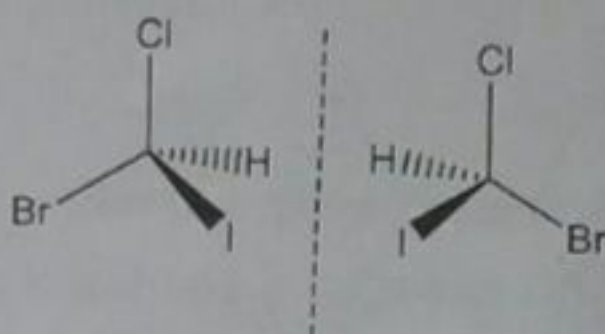
- 1.1 Build models of **cis-** and **trans-** isomers of 1,2-dichloroethene. Look up the melting and boiling points of these isomers. Give rationale for the differences.

3. Optical isomers

3.1 Build the models as shown for the two cases below. The broken line indicates an imaginary mirror. Convince yourself that in Case 1 the two structures are identical. That means the mirror images are super imposable on each other. If the two structures are not super imposable then they are different and are said to be optical isomers of each other.

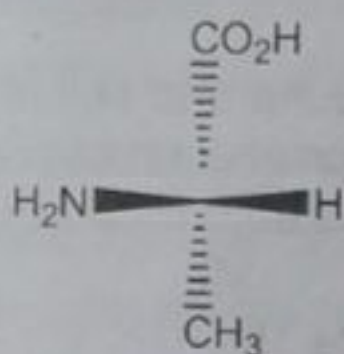


Case 1

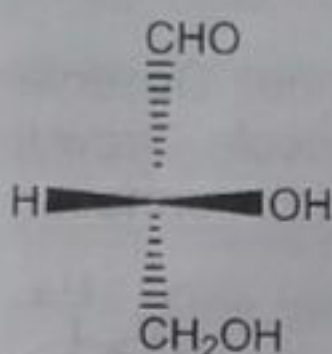


Case 2

3.2 Using the Cahn-Ingold-Prelog Convention specify the absolute stereochemistry at the chiral center of the following as either **R** or **S**:



L- Alanine



D-Glyceraldehyde

Quiz Questions

1. Write the mechanism for the conversion of maleic acid to maleic anhydride.
2. Write the 3D structures of cis- and trans – 1,4-cyclohexanediol.

Experiment 8

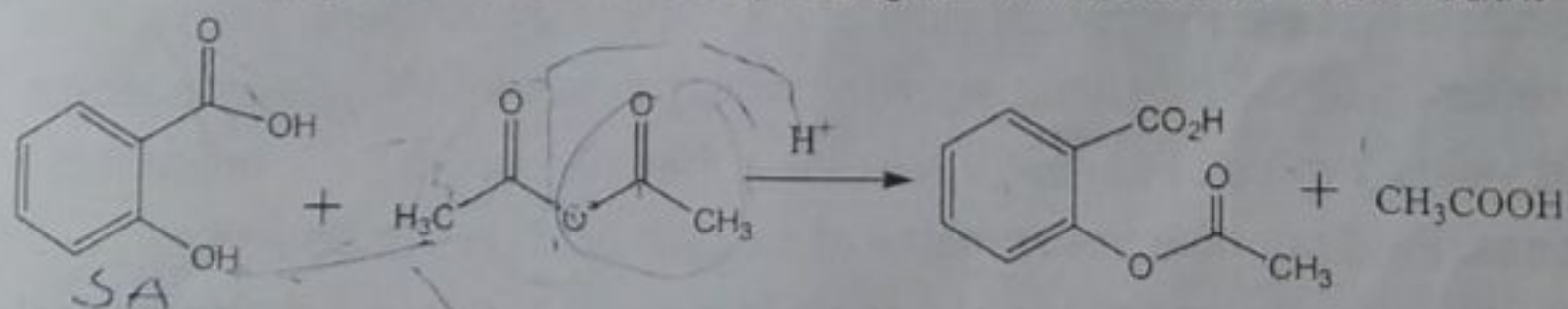
Preparation of Aspirin (Acetylsalicylic acid)

Objective

To prepare acetyl salicylic acid, commonly known as "Aspirin".

Theory

Acetyl salicylic acid (ASA), often sold under the name "Aspirin" is one of the most widely used painkillers (also known as analgesics) in the market. It is commonly recommended for the treatment of headache, cold, influenza, arthritis etc. The last synthetic step in the preparation of this drug is the subject of this experiment, which is the acetylation of salicylic acid using acetic anhydride in the presence of catalyst. It is interesting to note that the catalyst can be either a weak base (e.g. pyridine or sodium acetate) or a strong mineral acid such as sulfuric or phosphoric acid. To accelerate the rate of the reaction it is necessary to warm the reaction mixture using gentle heat such as placing the mixture over water bath.



The crude product can be isolated and recrystallized to give aspirin as a white crystalline solid, with a melting point of $134^{\circ}C$ - $135^{\circ}C$. Aspirin is stable in dry air but gradually hydrolyzes to salicylic and acetic acids in the presence of moisture.

Procedure

Place 1 g of salicylic acid (**SA**) into a clean test tube. The test tube has to

be dry because presence of even trace amount of water will spoil the experiment. Then add to it about 2 mL of acetic anhydride followed by 5 drops of concentrated sulfuric acid.

[Caution: Concentrated acids are very hazardous and extra care should be taken when handling to avoid contact and spill].

Warm the test tube by immersing it partially in a hot water bath for about 10 minutes. Then pour the mixture to a beaker or an Erlenmeyer flask that contains 30 ml of water. Place the beaker or Erlenmeyer flask in an ice bath. Crystals of aspirin will slowly precipitate. Collect the product by means of Suction Filtration. Remove the filter paper carefully with the product and place it in a paper box. Write your name on the paper box, and leave it in the hood till next week.

When you come back next week: Weigh the filter paper plus the crystals, subtract the weight of the filter paper from this value in order to determine the **yield of your product**.

Calculate first the theoretical yield of ASA in this experiment. This is done by referring to the balanced equation which shows that 1 mole of salicylic acid yields 1 mole of the ester assuming that the anhydride is present in excess. Find out how many moles of salicylic acid was used and determine how many moles of Aspirin this would result in (i.e. **theoretical yield**). Dividing your product yield above by the theoretical yield and multiplying it with 100 gives the **percentage yield**. For instance if the actual yield is half of the theoretical yield, the % yield is 50%. Note in your note book factors that contribute to decrease the actual yield.

Perform the following tests with the product:

1. Solubility test: Check the solubility of your product in water, ethanol and 10% sodium bicarbonate solution. In each case take small amount of

the dry crystals on the tip of a spatula and put it in a small test tube containing 1 mL or so of the test solvent. Repeat the experiment using salicylic acid. Explain the results.

2. Test for the phenolic hydroxyl group: To each of the above test tubes to which you have added aspirin and salicylic acid, add one drop of 1% ferric chloride solution. Record your observations and explain.

3 Determination of melting point: In the case of ASA its mp is 135-136°C.

Submit to your instructor what remains from your product.

Quiz Questions

1. Why is the carboxyl group of salicylic acid not acetylated?
2. What is the role of the acid catalyst? Also a base like pyridine can be used for the catalysis. Explain role of the catalyst by giving the mechanism of the reaction.
3. The melting point of salicylic acid is 157-159°C while that of aspirin, which has a higher molecular weight is around 135°C. Explain why?
4. What is the rational for the observed color reaction between salicylic acid and ferric chloride?



SA + AA
ASA FeCl₃
Color SA no color no SA

Spectroscopic analysis of ASA: A more powerful way of determining structure of organic compounds and also establishing their purity is to measure their proton and carbon spectra using nuclear magnetic resonance spectroscopy (NMR).

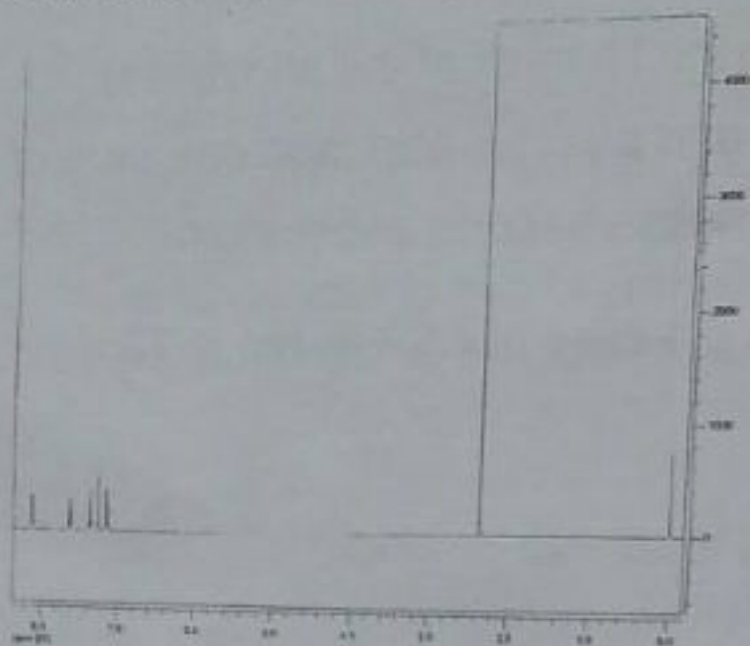


Fig. 9: Proton NMR spectrum of ASA

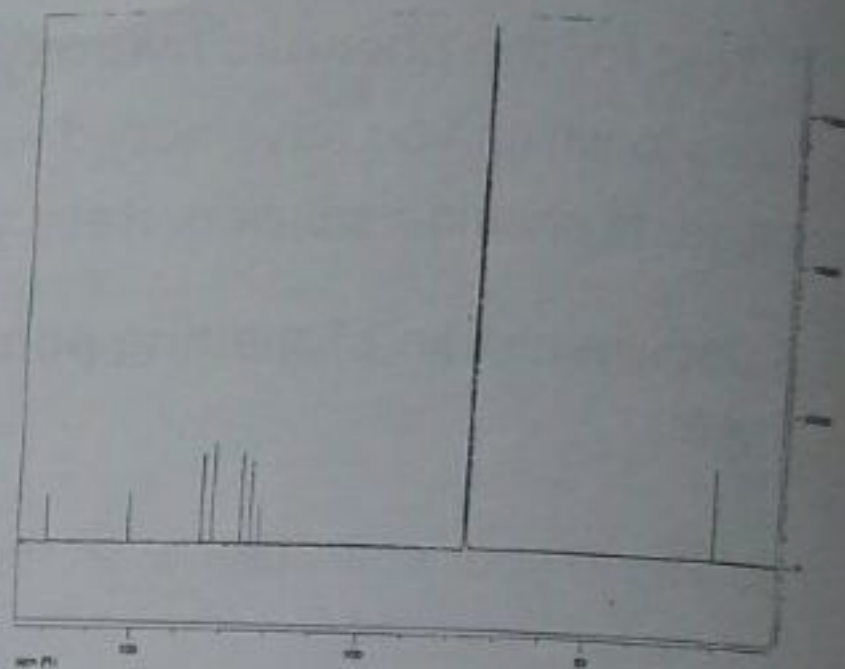


Fig. 10: Carbon- spectrum of ASA

Another useful technique is measurement of infrared (IR) spectra. As you will note below there is only one carbonyl peak at 1659 cm^{-1} in the spectrum of Salycilic Acid (SA). However in the IR spectrum of ASA there are two such peaks one for the carboxyl carbonyl (1691 cm^{-1}) and another one at 1753 cm^{-1} due to the acetyl carbonyl.

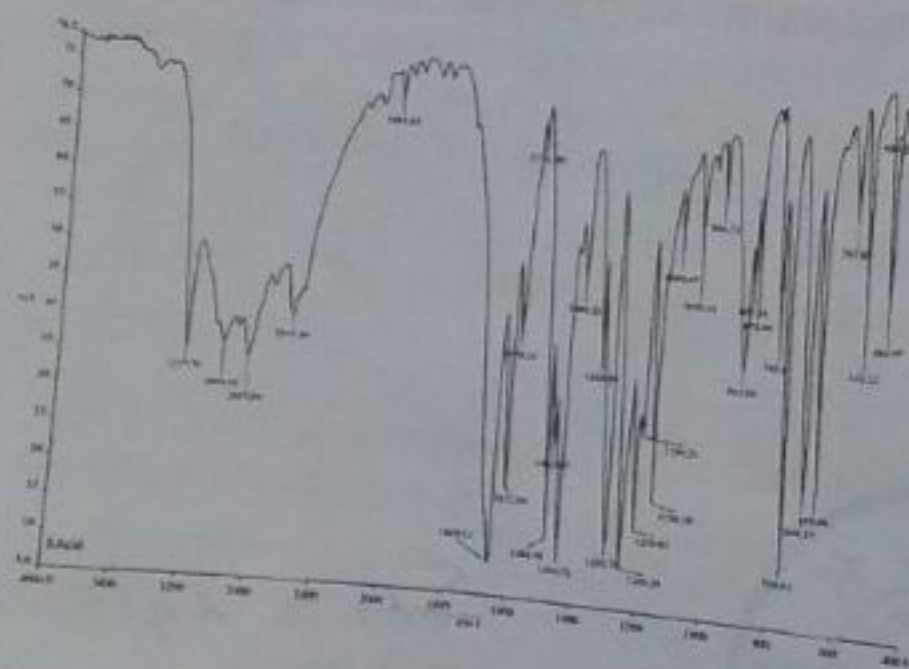


Fig. 11: Infra red spectrum of SA

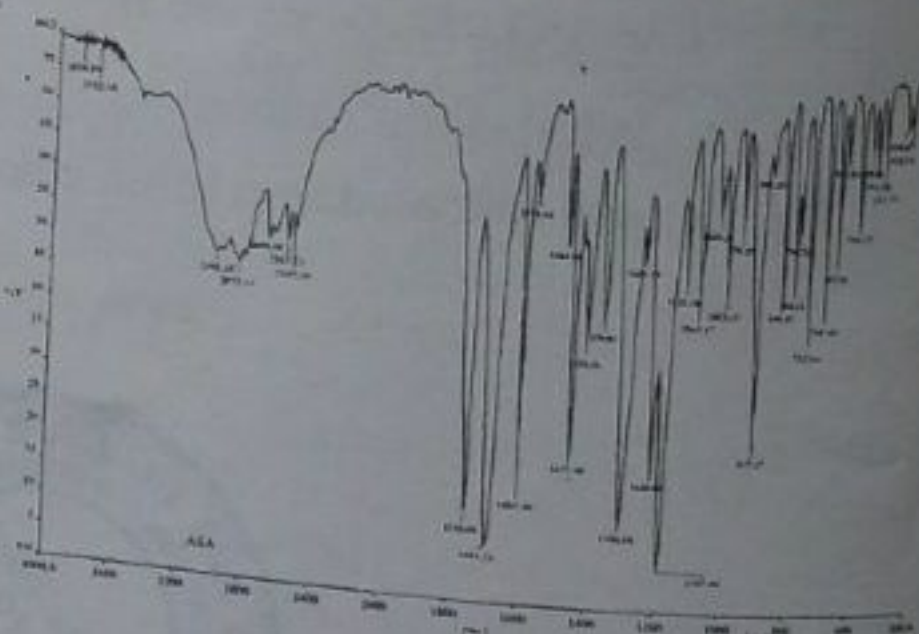


Fig. 12: IR spectrum of ASA

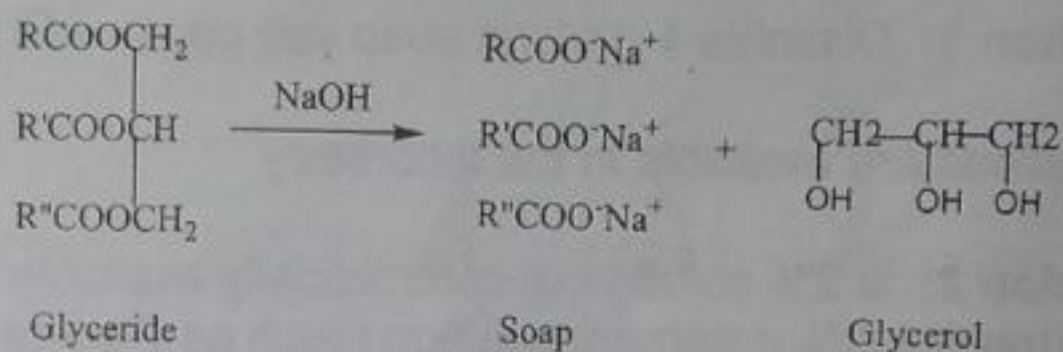
Experiment 9

Preparation of Soap

Objective: To prepare ordinary soap and to examine its properties.

Theory

The art of making soap has been known since ancient times and is still used today industrially. Fats are esters of fatty acids and glycerol. That is why fats are sometimes referred to as glycerides. When these esters are hydrolysed by use of base such as sodium hydroxide they give rise to sodium salt of long chain fatty acids and glycerol. The process of converting fats to salt of fatty acids and glycerol is known as saponification. Thus the hydrolysis of fat with alkali yields glycerol and three molecules of the salt of the fatty acid (soap).



- ① The most widely occurring esters in nature are fats from animal and vegetable origin. Animal tallow, lard, palm oil, coconut oil, olive oil, peanut oil, cotton seed oil, corn oil etc. are but a few of the naturally occurring glycerides.
- ② Natural fats and oils are mixtures of glycerides. For instance, tallow contains myristic, palmitic, stearic, oleic and linoleic esters of glycerol. The difference between fats and oils is caused by the fact that fats are composed of glycerides of saturated acids, while oils are made up of unsaturated fatty acids (caused by the presence of double bond). The extent of unsaturation in fat or oil is expressed in terms of its **iodine value**, which is defined as the number of grams of iodine which will add to 100 g of fat or oil. Thus a high iodine value as in vegetable oil indicates a high degree of unsaturation.

The soap molecules possess detergent or cleansing properties because of their ability to form aggregates with fat soluble materials in which the long, fatty acid chains surround the "dirt" in such a way as to enclose it within a cluster of soap anions leaving the hydrophilic carboxyl groups as a peripheral water solubilizing envelope. The resulting "Dirt + soap" complex can then be carried away by the water.

Procedure

Note how a reflux set up is arranged by observing the set up prepared by your instructor and arrange your own reflux set up. In a round bottom flask place 10 g of beef tallow, 20 mL of 20% sodium hydroxide solution, 15 mL 95% ethanol and 2 boiling chips. Ethanol helps the fat to dissolve easily. Light the Bunsen burner

using gentle flame and reflux for about 1 h. A homogenous soap solution now forms.

Turn the heat off, wait for a few minutes to let the flask to cool a bit, and then pour the warm solution into a beaker containing 100 mL of **brine solution** (i.e. saturated salt solution). Cool the beaker in an ice bath. Separate the solid soap from the mixture by decanting the liquid or filtering using suction filtration. Transfer the solid soap to a small beaker and place the beaker over a wire gauze and heat it till the soap becomes soft. Turn the heat off, wait for a few minutes so that the beaker cools down to room temperature and then put the beaker in an ice bath so that the soap becomes harder.

Remove the soap from the beaker using spatula and conduct the following tests:

Test Solution 1: Dissolve 1 g of the soap you prepared in 50 mL of water

The following solutions are available in the laboratory

Test solution 2: is 2% solution of commercially available ordinary soap.

Test solution 3 is 2% detergent solution (such as Omo or Roll etc)

Perform the following experiments on the above three test solutions and note the results in your report book.

1. Test the alkalinity of each of the above solutions by means of litmus paper.
2. Place 2 mL of **Test Solution 1**, into 4 test tubes. Add 3-5 drops of calcium chloride solution to the first test tube, 5% magnesium chloride to the second, 5% ferric chloride to the third and 6M hydrochloric acid to the fourth. Shake each test tube well and note your observations and present the result in a Table.

Repeat this experiment with Test Solutions 2 and 3.

Explain the different behaviors of the test solutions in the Discussion section of your report.



Fig. 13: Reflux set up

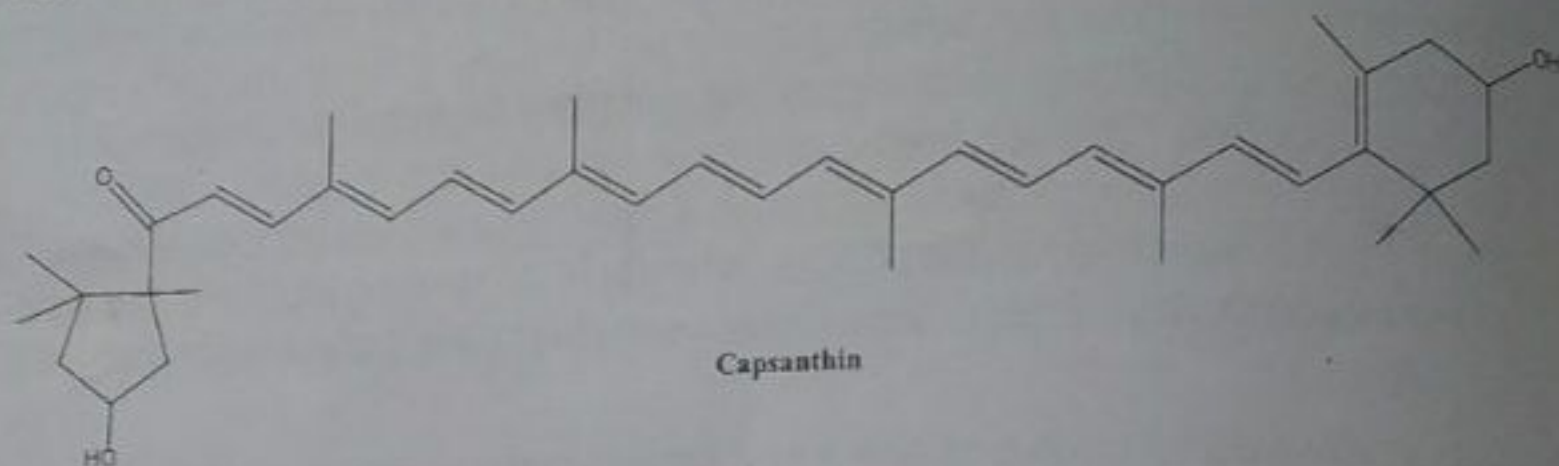
Quiz Questions

1. Why is glycerol very soluble in water?
2. What is the mechanism by which soap removes dirt?
3. Which part of soap molecule associates with "dirt" and which part with water?
4. Why is better to use a reflux set up for a reaction that needs the solution to be boiled for a long period?

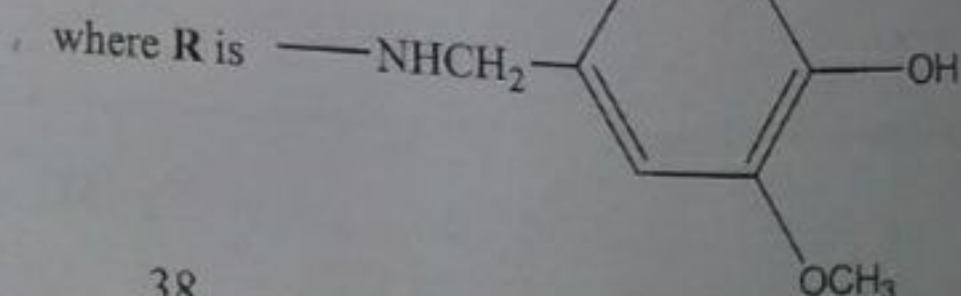
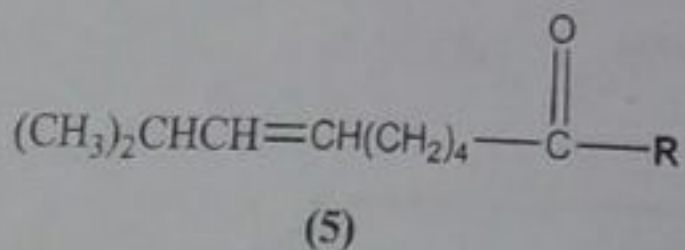
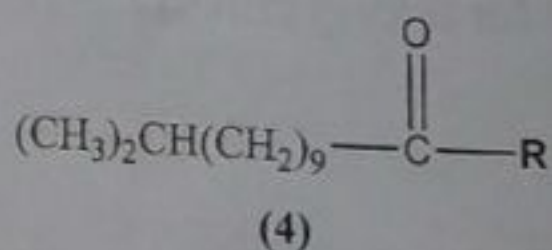
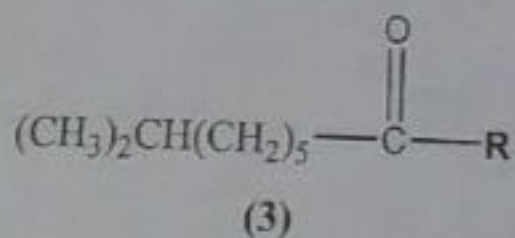
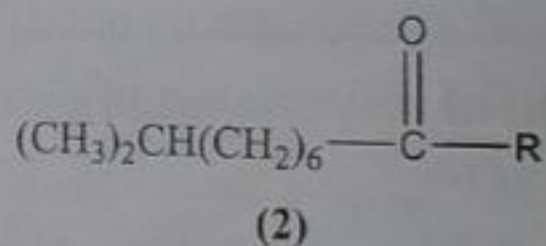
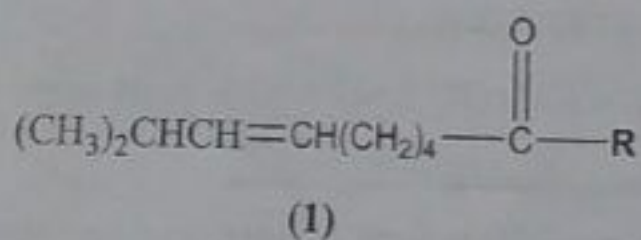
Capsicum annum or Berbere

Columbus, on his discovery of the New World found capsicum (chillies), which is America's most important contribution to the spices. Capsicums soon spread throughout the old world. That is how we got the Berbere that we very much like to spice our food with. If the powdered chilli is extracted with hexane the colourful components known as capsanthins are extracted.

The structure of the major pigment of **capsanthin** is shown below.



The pungent or hot components known as **capsaicins** are extracted with the more polar solvent alcohol. These are vanillyl amides of differing side chain: capsaicin (1), dihydrocapsaicin (2), nordihydrocapsaicin (3), homodihydrocapsaicin (4), homocapsaicin (5), the major one being **capsaicin**.



Experiment 10

Chromatography

Objective

How best to use chromatographic techniques for separation and identification of organic compounds.

Theory

In 1903, the Russian botanist Tswett passed extracts of plants through a column filled with chalk powder. He succeeded to separate the pigments present in the extract. He coined the term **chromatography** from the Latin for "**color writing**"). This method was further improved when Martin and Synge in 1940, managed to separate a mixture containing amino acids by using a silica gel column. In 1944 Martin (with his colleagues R. Consden and A. H. Gordon) created **paper chromatography**. These scientists were given the Nobel Prize in 1952.

Chromatography is one of the most valuable techniques for the separation, identification and purification of organic compounds. Chromatographic separations depend on the differences in the partition coefficients of the components between two immiscible phases: a stationary phase and a mobile phase. The mobile phase can be liquid or gas, while the stationary phase may be liquid or solid. In **partition** chromatography such as liquid-liquid chromatography, components partition between two liquid phases, while in gas liquid chromatography (GLC), the mobile phase is gas while the stationary phase is liquid.

Mobile Phase	Stationary Phase	Technique
Vapour	Solid	Gas chromatography (Gas-solid chromatography)
	liquid	Gas chromatography (gas-liquid chromatography)
Liquid	Solid	Adsorption chromatography (liquid-solid chromatography)
	Liquid	Liquid-liquid partition (liquid-liquid chromatography)

The way chromatography is practiced is by subjecting a sample, which is usually a mixture of two or more substances, to a chromatographic process. A mobile

phase is allowed to flush the components through the system. The different components of the mixture move at different rates. The smaller the affinity a molecule has for the stationary phase, the faster it comes out of the process. These differential rates of migration as the mixture moves over adsorptive material provide separation. Repeated sorption / desorption processes that take place during the movement of the sample over the stationary bed leads to separation of the components of the mixture.

Paper Chromatography

In paper chromatography, a special paper, which resembles filter paper is used. The paper acts as a flat "column," providing the solid phase on which a stationary polar solvent and a mobile, less polar solvent could interact. Paper chromatography is an example of liquid-liquid chromatography (LLC). It finds application in the separation and identification of small amounts of materials.

The cellulose, of which ordinary filter paper is composed, has water absorbed on its surface and inside. The water is called the stationary phase and the cellulose fiber is the solid support. A material applied on the filter paper when immersed in a solvent moves due to capillary action and partitions between the mobile solvent and the cellulose bound water. Hence the name liquid-liquid chromatography and this is an excellent example of **partition chromatography**.

The technique of paper chromatography is quite simple. The solution of the sample is applied on a tiny spot at one end of the paper strip by means of a capillary tube. The end of the paper nearest the spot is immersed in a suitable solvent system, without immersing the spot itself. The solvent rises up the paper by means of capillarity.

The substances applied on the spot move at different rates based on the distribution of each component between the stationary and mobile phases. The distance each component travels, relative to the origin, is measured using a cm ruler. Also to be measured is the distance the solvent traveled. Dividing the distance each substance traveled by the distance the solvent traveled gives what is known as **R_f (retention factor or value)** of each component. The R_f is dependent on factors such as nature of the component, type of paper used and the solvent system.

Other types of chromatography

- Thin layer chromatography
- Column Chromatography
- High-pressure liquid chromatography (HPLC).
- Ion-exchange chromatography (IEC) and ion chromatography (IC).

- Affinity chromatography
- Electrophoresis.

procedure

a. **Paper chromatography (PC):** Your instructor will give you a **mixture** containing any two of the following dyes: eosin (**EO**), fluorescein (**FI**), malachite green (**MG**) and methyl orange (**Mo**). In the middle of a strip of Whatman Chromatography Paper size 6 x 16 cm and 2 cm from the bottom, spot by means of a capillary tube a solution of **the mixture** assigned to your group. Circle with pencil (**caution!** not ink) the spot and write Mix at the top middle of the paper directly above the circle. To the right and left of this spot and with a separation of at least 1 cm apply the reference substances. Mark the identity of each reference substance at the top of the paper using the following abbreviations: eosin (**EO**), fluorescein (**FI**), malachite green (**MG**) and methyl orange (**Mo**).

Pour 35-40 mL of the developing solvent (n-butanol/ethanol/2N ammonia 2: 2: 1) into the provided chromatographic tank. Place a strip of ordinary filter paper on one side. This will help the space inside the jar to be saturated with the vapour of the solvent system. Place then your spotted chromatography paper in the jar and develop the chromatogram. The solvent will then gradually rise through the paper. Let this proceed till the solvent front has risen to approximately 4/5 of the height of the paper, remove the paper from the tank and immediately mark the solvent front with pencil. Allow the paper to dry and then measure the distance from the starting point to the center of each of the oval spots. Calculate the R_f value of each of the known components i.e. MG, FI, EO and Mo. Then calculate R_f of the spots that appear from the middle spot. Suggest what the composition is of the sample given to you by your instructor i.e. the one you applied in the middle spot.

b. **Thin layer Chromatography (TLC):** The use of TLC to detect components of extracts of **one** of the common spices (e.g. **chilies** or **Capsicum annum**, **Berberis**) will be demonstrated by your instructor. Your instructor will take 1 g of the spice and extract it first with hexane. Decant the hexane and label it as **hexane extract**. To the residue add few milliliters of ethanol, shake well and then decant to get the **ethanol extract**. On a small strip of TLC plate apply on the left side the hexane and on the right side the ethanol extract. Develop the plate first using as a solvent system n-hexane only. Take the TLC plate out of the chamber and put it in another chamber that contains acetone/hexane 1:1.

c. **Column chromatography (CC):** This demonstration is done by using a tiny column packed with silica gel to which one of the above extracts will be applied. Elution will be done first with hexane, and then with acetone. This will help the student appreciate the importance of CC to separate natural products.

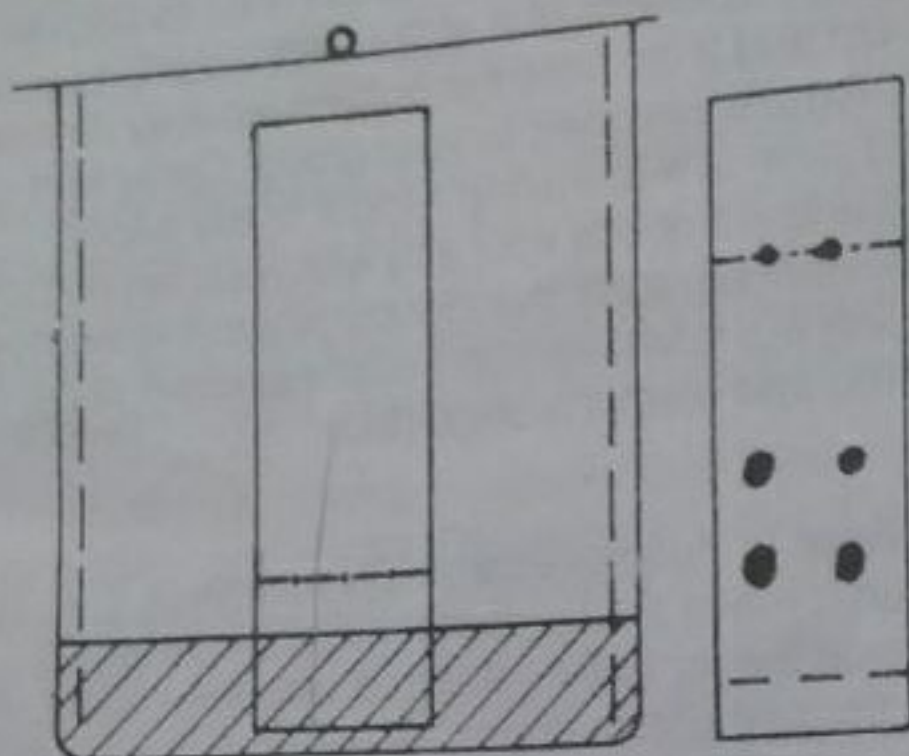


Fig. 14: TLC set up



Fig. 15: Column chromatography

Quiz Questions

1. On silica gel plate, polar compounds generally have lower R_f than non polar. Why?
2. Thin layer chromatography can also be used for final purification of small amounts of compounds. It then called Preparative TLC or simply PTLC. Discuss how PTLC can be important to separate mixtures of natural compounds.

Experiment 11

Introduction to Proteins

Objective

To learn how to test proteins by chemical means

Theory

Proteins are the most abundant organic molecules of living cells and serve as structural units of muscle, skin, hair and other tissues. Proteins are polymers with molecular weights ranging from 5000 to 1 million or more. On acid hydrolysis proteins yield a series of organic compounds of low molecular weights known as amino acids. Some amino acids are usually found as the building blocks of proteins. In protein molecules, the amino acid units are covalently bonded by amide linkages known as peptide bonds ($-\text{NHCO}-$). Repeating units of peptides are known as polypeptides. Proteins are important nutrients, which are obtained from several types of food. The digestion of proteins in the stomach gives rise to amino acids and these in turn provide the building blocks for the synthesis of proteins that we need to build and maintain our body.

Human beings are not capable of synthesizing certain amino acids and must therefore obtain these in one form or another in the diet. Such amino acids are known as essential amino acids.

The presence of the amino and carboxyl functional groups in proteins makes these substances amphoteric (i.e. act as acids or bases). Proteins generally coagulate and harden when heated or treated with acids. The protein is then said to be denatured. The hardening is due to changes in the stereochemistry of the constituent protein molecules.

Treatment of protein with concentrated nitric acid results in the development of yellow colour. This test which is known as the xanthoproteic test is based on the nitration of benzene ring of certain amino acids such as tyrosine, phenylalanine, or tryptophan.

Proteins containing free amino groups react with nitrous acid and liberate nitrogen gas.

The principal protein in milk is **casein** and it is the chief ingredient of cheese. Cow's milk contains about 3% casein. Casein has an approximate molecular weight of 375,000 and is made up of at least 16 amino acids.

Procedure

Isolation of Casein

Place 25 mL of non fat milk in an Erlenmeyer flask, place the flask in a water bath and warm it to about 50°C . Add 15 drops of glacial acetic acid to the warm milk. A white precipitate is formed, which is mainly casein. Filter by suction. Wash the casein with 3 mL of water and then with 3 mL alcohol. Divide the product into four portions and place each on dry piece of paper and perform the following tests:

Tests on Casein

- Solubility tests:** Add the first portion of casein to a test tube containing 3 mL of concentrated hydrochloric acid. Shake well, warm on the water bath. Describe the test and its significance. Repeat the solubility test using the second portion in 1 mL of 20% NaOH.
- Xanthoproteic test:** Add the third portion to a test tube containing 10 drops of concentrated nitric acid. Observe and note the changes that take place. Explain the significance of this test with respect to amino acid composition of casein.
- Nitrous acid-casein Reaction:** The fourth portion should be added to a test tube containing 2 mL of 10% hydrochloric acid. Cool the mixture by placing the test tube in an ice bath. Now add 1 mL of 5% sodium nitrite (NaNO_2) solution. Keep the test tube in the ice bath and note if gas bubbles appear. What are these bubbles? Explain by means of a chemical equation how these bubbles arise.

Tests on Egg Albumin:

Obtain about 10 mL of the white of an egg and divide it equally into four portions and conduct the following test:-

- Xanthoproteic Test:** Pour the first portion of the albumin into a test tube containing 5 mL of water. Shake well and add 10-15 drops of conc. nitric acid. Warm gently and note the colour change. Cool the solution and neutralise it with 5 mL of 10% sodium hydroxide. Record your observations in your notebook.
- Test for Sulfur:** Put 1 mL of albumin in a test and add to it 7 mL of 10% sodium hydroxide solution. Place the test tube for 15 min in hot water bath. Cool the solution and acidify it by adding the provided HCl solution drop wise till it turns blue litmus paper to red and then immerse the test tube in hot water bath. Bring a piece of moist lead acetate paper to the mouth of the test tube and observe change in its colour. Note and explain your observations.

Benedict's Test:- As above but this time you use Benedict's Solution instead of Fehling's solution. A change in color from blue to yellow and then red indicates the aldehyde group. This test is also positive for glucose and lactose but negative for sucrose.

Osazone Formation: Monosaccharides and reducing disaccharides react with phenylhydrazine to yield osazones. (Write the equation). Osazones are crystalline derivatives of sugars, which are extremely useful in the identification, and characterization of sugars. Osazone derivatives of simple sugars are formed as follows:

Bring a water bath to boiling. Add 2 mL of phenylhydrazine reagent and 3 mL of 5% glucose solution to a test tube. Mix well by shaking. Place the test tube on a water bath and note if osazone is formed as a precipitate in 3-5 minutes. Repeat the test using 5% sucrose solution. Explain why glucose gives positive test and sucrose negative.

Quiz Questions

1. Show by chemical equations formation of a dipeptide from glycine gly-gly and the tripeptide cys-ala-gly? Get the structure of the amino acids from your textbook.
2. Why does the casein precipitate out after addition of glacial acetic acid?
3. What is the basis for the xanthoproteic and the Biuret tests?
4. The properties of glucose is better accounted for by the cyclic hemiacetal and it has been shown that 90% of glucose exist in this form. Draw the hemiacetal cyclic structure formed between the aldehyde group at carbon 1 and the hydroxyl group at carbon 5?
5. Write the structures of sucrose and lactose, and show how one might distinguish between the two in the laboratory?
6. What is a Fehling's solution? Why is this test very useful in clinical chemistry?
7. Write the equation for the reaction between phenylhydrazine and glucose? What is the mechanism of the reaction?

Experiment 12

Carbohydrates

Objective

To examine the chemical properties of carbohydrates

Theory

There is probably more carbohydrate in the biosphere than all other organic matter combined. This is mainly because of the extraordinary abundance of two polymers of D-glucose namely cellulose and starch. Monosaccharides, or simple sugars, are made up of a single polyhydroxy aldehyde or ketone unit. The most abundant monosaccharide is the simple six-carbon sugar D-glucose. Other examples include: glyceraldehyde, erythrose, ribose, mannose, fructose etc. Sugars that have three carbon atoms in the chain are called trioses and those with four, five, six and seven carbons are tetroses, pentoses, hexoses and heptoses respectively. Furthermore, if the carbonyl group is at the end of the chain, the monosaccharide is called an aldose, where as if it is at any other position the sugar is a ketose. Pentoses and hexoses mainly exist in cyclic forms.

Oligosaccharides may be illustrated by common disaccharides like lactose (found in milk), sucrose (in cane sugar), cellobiose (in cellulose) and maltose (product of action of enzyme amylase on starch).

Procedure

Solubility tests: Using the tip of spatula transfer into three test tubes small amounts of glucose, sucrose and starch, determine solubility in water (2 mL). Repeat the experiment using ethanol (2 mL). Tabulate your results.

Using the provided 5% aqueous solution of glucose and 5% sucrose conduct the following tests:-

The Molisch Test: Add 2 mL of concentrated sulfuric acid into a test tube. In another test tube place 2 mL 5% aqueous solution of glucose to which 2 drops of the **Molisch reagent** (15% solution of alpha-naphthol in ethanol) should be added. Pour gently the sulfuric acid down the inside of the test tube containing the glucose solution. Hold the latter test tube at an angle of about 30° . Do not shake the test tube. Two separate layers are formed. The purple coloured ring formed at the junction of the two liquids is a positive test for carbohydrates.

Fehling's Test for reducing Sugars: Place 3 mL of Fehling's solution in a test tube and heat it on water bath. Add drop wise 5% glucose solution. Observe any colour change. Continue adding the solution till the original blue color of the Fehling's solution disappears. Why? Repeat the test with sucrose solution and record your results. Explain your results.

Benedict's Test:- As above but this time you use Benedict's Solution instead of Fehling's solution. A change in color from blue to yellow and then red indicates the aldehyde group. This test is also positive for glucose and lactose but negative for sucrose.

Osazone Formation: Monosaccharides and reducing disaccharides react with phenylhydrazine to yield osazones. (Write the equation). Osazones are crystalline derivatives of sugars, which are extremely useful in the identification, and characterization of sugars. Osazone derivatives of simple sugars are formed as follows:

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Quiz Questions

1. Show by chemical equations formation of a dipeptide from glycine gly-gly and the tripeptide cys-ala-gly? Get the structure of the amino acids from your textbook.
2. Why does the casein precipitate out after addition of glacial acetic acid?
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4. The properties of glucose is better accounted for by the cyclic hemiacetal and it has been shown that 90% of glucose exist in this form. Draw the hemiacetal cyclic structure formed between the aldehyde group at carbon 1 and the hydroxyl group at carbon 5?
5. Write the structures of sucrose and lactose, and show how one might distinguish between the two in the laboratory?
6. What is a Fehling's solution? Why is this test very useful in clinical chemistry?
7. Write the equation for the reaction between phenylhydrazine and glucose? What is the mechanism of the reaction?

Experiment 13

Qualitative Organic Analysis

Objective: To identify unknown organic compounds by solubility classification and chemical tests.

Theory

The characterization of compounds is an important aspect of the work of a chemist. Some of the compounds that need to be characterized may be totally unknown. If this is the case extensive research may be required to completely characterize the compound and determine its correct structure. In most cases, however, a chemist is confronted with the problem of identifying compounds from among those already known. A systematic approach to the analysis of chemical compounds is therefore highly desirable. Such an approach helps one to exclude as many structural possibilities as possible, gradually narrowing down the structure of the unknown step by step to few compounds and finally selecting one of the proposed structures as the most likely structure for the unknown. In this Experiment attempts will be made to introduce the student to the systematic identification of organic compounds.

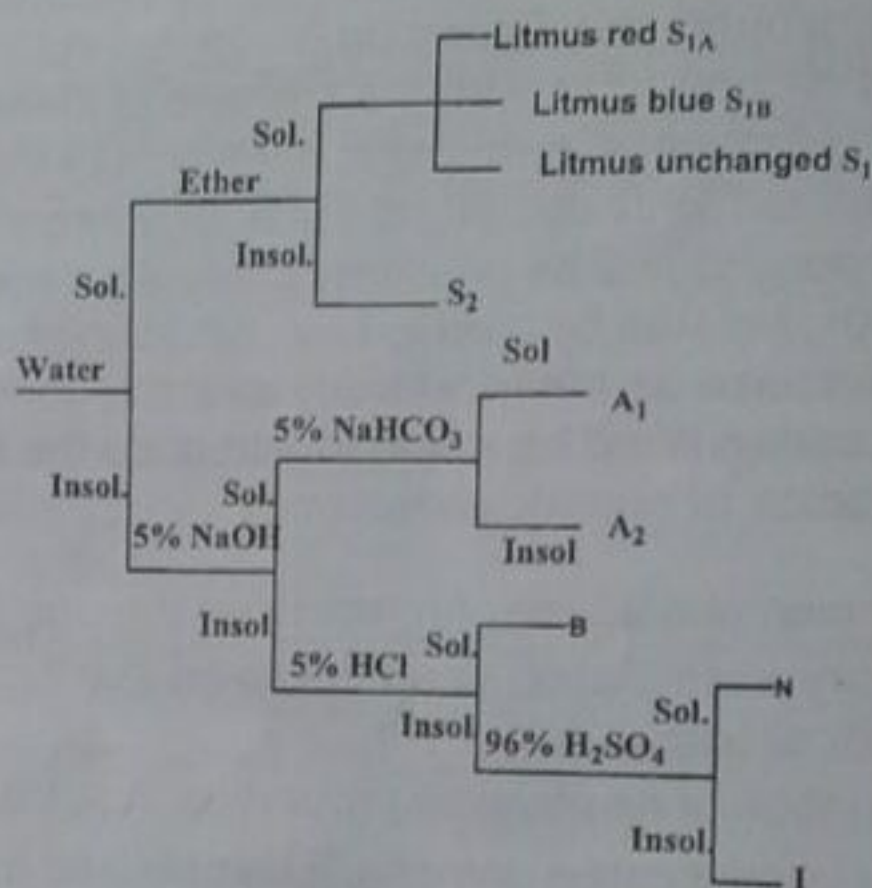
To identify an unknown substance one has to go through the following steps:

- Preliminary examination of an unknown substance: is it liquid or solid? appearance, colour, odour etc
- Determination of its physical properties: mp, bp, refractive index, R_f .
- Solubility classification: What is it soluble and insoluble in and relating this to the type of functional group present in the unknown.
- Functional group identification by chemical tests
- Preparation of derivatives and listing possible structures
- Final selection of the most likely structure for the unknown

Preliminary examination: Knowing the physical state of a given substance, i.e. whether it is a liquid or solid, its colour and odour, aids greatly in the identification process. A simple but useful experiment in some cases is the **ignition test** performed on a small amount of the substance. By this method it is possible to distinguish aromatic hydrocarbons from those that are aliphatic. In general aromatic compounds e.g. benzene generally burn with sooty flame, while aliphatic compounds e.g. butane (as in *butagas*) burn with clear bluish flame. Sugars char and organic salts have residual ash. Thin layer and paper chromatographic methods also serve as one of the most rapid techniques in the identification of unknown substances.

The determination of physical properties: Physical constants such as melting and boiling points, specific gravity and refractive index are essential data that aid during identification of unknown substances.

Solubility classification: This is a very simple test but one that leads to very useful clues in identification. Solubility of an organic compound in a solvent may be due to a chemical reaction as when an organic acid dissolves in an aqueous basic solution or may just be caused by simple miscibility as when benzoic acid dissolves in ethanol. From a solubility experiment the presence and type of functional group may sometimes be determined. Thus the solubility in 5% NaOH of a water insoluble unknown is an indication of the presence of an acidic functionality e.g. $-\text{COOH}$ and phenol $-\text{OH}$ groups. These two can be further distinguished by testing with dilute NaHCO_3 , which dissolves carboxylic acids but not phenols. Basic compounds are differentiated by the ease with which they dissolve in 5% HCl. One can classify organic compounds using the chart below:



- S₂** Salts of organic acids (RCO_2Na , RSO_3Na), amine hydrochlorides (RNH_3Cl), amino acids, carbohydrates.
- S_A** Monofunctional carboxylic acids with five or less carbon atoms
- S_B** Monofunctional amines with six carbons or fewer.
- S₁** Monofunctional alcohols, aldehydes, ketones, esters, nitriles, and amides, that have five carbons or fewer.
- A₁** Carboxylic acids with more than six carbons; phenols with electron withdrawing groups.
- A₂** Weak organic acids such as phenols, enols, oximes, imides, sulfonamides, with more than five carbons.
- B** Aliphatic amines with eight or more carbons, anilines.
- MN** Misc compounds containing nitrogen or sulfur with more than 5 C atoms.
- N₁** Alcohols, aldehydes, ketones, esters + 1 funct. group and 5- 9 C atoms
- N₂** Alkenes alkynes, ethers, some aromatic compounds
- I** Saturated hydrocarbons, haloalkanes, aryl halides, diaryl ethers, deactivated aromatic compounds.

Detection of functional groups by chemical tests: It is well known that a functional group is responsible for the physical and chemical properties of a compound. Hence one of the most useful information in determining the identity of an unknown is to determine the nature of the functional group(s) present in the molecule. This can be done by a variety of tests. For instance presence of **unsaturation** in a compound can be confirmed by the **Potassium Permanganate Test** by dissolving the unknown in water or acetone (1 mL) and adding three drops of aqueous potassium permanganate solution. Disappearance of the purple colour with the formation of a brown solution is indicative of the presence of a double bond in the compound.

For amino acids, Ninhydrin Test: Place one drop of the aqueous solution of the amino acid on ninhydrin paper (prepared by soaking filter paper in 0.2% aqueous solution of ninhydrin followed by drying). Blue or violet spot or ring is considered a positive reaction.

Copper Sulphate test: Add an aqueous solution of the compound to an aqueous copper sulphate solution. A deep blue colouration is due to a complex formation with aliphatic amino acids.

Aldehydes and ketones, 2,4-Dinitrophenylhydrazine (DNP) Test: Place in a test tube a small amount of the compound to be tested using tip of the spatula. Add to it 2 mL of water. If the substance is not readily soluble in water add a few drops of ethanol. Add 2 mL of the reagent and shake well. A yellow or orange precipitate indicates presence of a carbonyl group.

Test for carbohydrates, Fehling's test: To a small amount of the sample placed in a test tube, add 2 mL of water and 1 mL of Fehling's solution. Then heat the mixture on water bath. The formation of red precipitate indicates monosaccharide as well as reducing disaccharide like lactose. Hence this test is positive for glucose and lactose but negative for sucrose. Here is another alternative test for sugars:

Benedict's Test: Use instead 2 mL of Benedict's solution and heat the mixture on the water bath. A change in colour from blue to yellow and then red indicates presence of an aldehyde group. This test is also positive for glucose and lactose but negative for sucrose.

Test for carboxylic acids and phenols:

Litmus paper test: Bring a piece of moist blue litmus paper in contact with the solution of the compound. If it turns red the presence of carboxylic acid is indicated.

Sodium bicarbonate test: Take about 2 mL of cold saturated sodium bicarbonate solution in a test tube and add a small quantity to your compound. Effervescence due to the evolution of carbon dioxide indicates acid functionality. Phenols give negative result in this test.

Ferric chloride test for phenols: To the compound placed in the test tube, add 2 mL of water, shake well, then add 2 drops of 1% neutral ferric chloride solution. Phenolic compounds usually give violet colour.

Tests for primary aromatic amines: To the substance in a test tube add 2 mL dilute hydrochloric acid (5%). If the substance is insoluble add 3-4 mL of water. Cool the test tube by placing it in an ice bath and add to it 2 mL of pre-cooled 2% sodium nitrite (NaNO_2) solution. Add 1 mL of this solution to 1% beta-naphthol in 5% NaOH. Orange red colored dye confirms primary aromatic amine.

Procedure

Perform the following:

1. **On known compounds:** From the following list select at least three compounds and perform on each classification and identification tests.

Neutral compounds: Acetanilide, benzophenone,

Acids and phenols: Benzoic acid, phenol

Basic compounds: Aniline, toluidine.

Salts: Sodium acetate

Carbohydrates: Glucose, sucrose

Amino acids: Glycine, alanine

2. **On an unknown compound:** Your instructor will assign to you an unknown compound that you should identify. Write in your note book the code of the samples that you are given. You are then expected to note its physical properties (colour, odour, solid or liquid etc), then determine its solubility class, and perform appropriate chemical tests. Based on your observations you should then report the identify of your unknown. Write your arguments properly in your note book.

Quiz questions

1. Write chemical equations for the following tests: Fehling's test; 2,4-DNPH test of carbonyl compounds; reaction of primary aromatic amines with nitrous acid and the coupling reaction with alpha-naphthol
2. Why are phenolic compounds generally insoluble in sodium bicarbonate but soluble in sodium carbonate solutions.
3. Why does our fingers give positive reaction with ninhydrin solution?

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3. Why does our fingers give positive reaction with ninhydrin solution?

Appendix

NOTES TO INSTRUCTORS AND TECHNICIANS

Experiment 1: Contaminated benzoic acid and salicylic acid are recommended. Students have to be shown how to handle the Bunsen burner. They should be told not to light their burner by using lighter or matches. They have to also be shown how to heat their filter funnels using a Bunsen flame before attempting to filter the hot saturated solution.

Experiment 2: The main points of how to work safely in the laboratory should be covered.

Experiment 3: For the simple distillation, ethanol contaminated with a dye such as congo red is used.

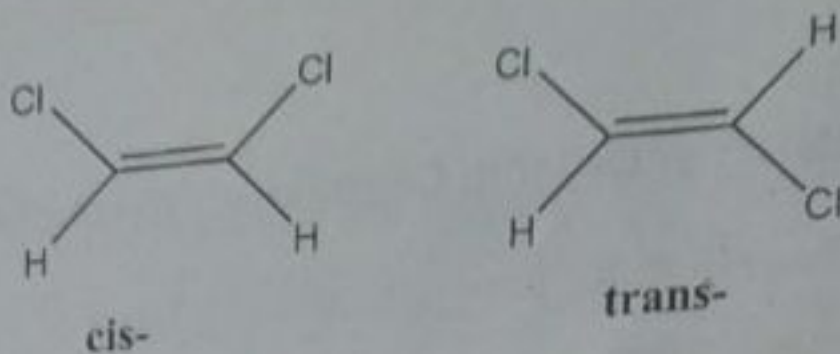
Experiment 4: The fractional distillation could be done by using 1:1 mixture of ethanol and water.

Experiment 5: You may use eucalyptus oil for the Experiment B. Get copy of gas chromatograph of this oil. Post it on the notice board in the laboratory so that the students appreciate use of GC to analyse volatile oils. Mark the highest peak as due to 1,8-cineole, the major substance (>70%).

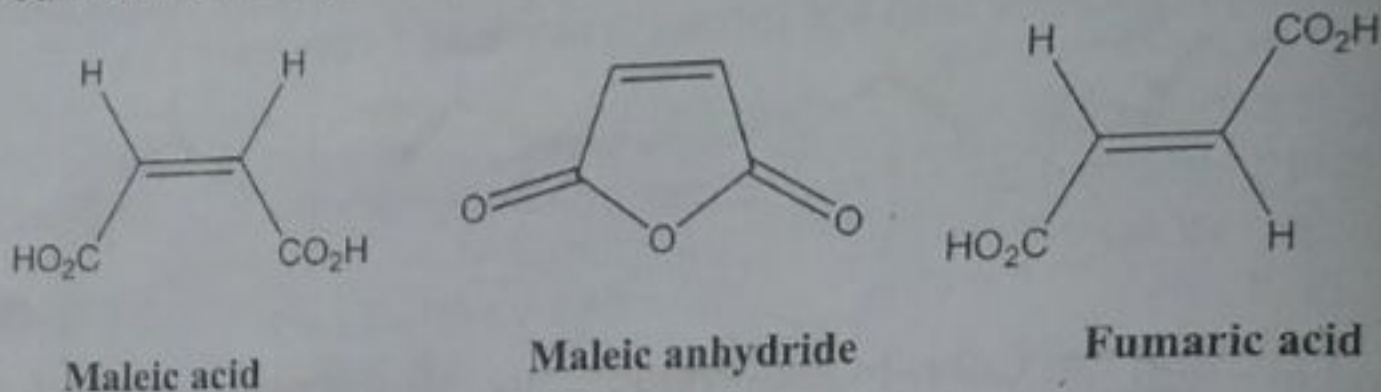
Experiment 6: It should be noted that gasoline gives positive reaction for the unsaturation test. The gasoline should be used as purchased from a petrol station. The result of the gasoline test should be compared with hexane. The acetylene generation needs to be well organized by the instructor especially when the lab group is large, so that small enough groups of students can observe the reactions at anyone time.

Experiment 7: It is possible here for students to write their lab reports during the session by drawing diagrams directly from models. It is useful to always build both enantiomers of the chiral molecules in order to see clearly their non-superimposability. A quiz at the beginning of this experiment based on stereochemical terms is recommended.

Experiment 8: A perennial problem in this experiment is the observation that the product may in some cases give rise to positive ferric chloride test, showing presence of trace of SA. This is Ok. The students should leave their products to dry over the week and should then weigh, determine mp and calculate the yield. A quiz question about yield calculation should be included in this experiment.

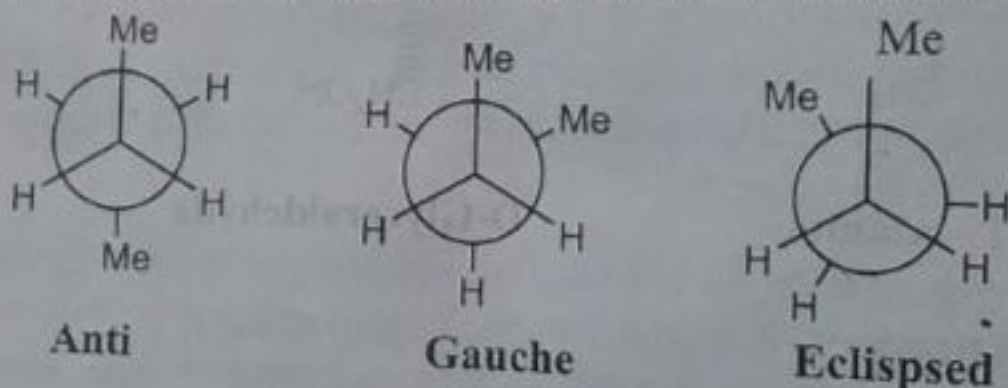


1.2 Build models of maleic and fumaric acids. Why is it that maleic acid can form an anhydride but not fumaric acid?

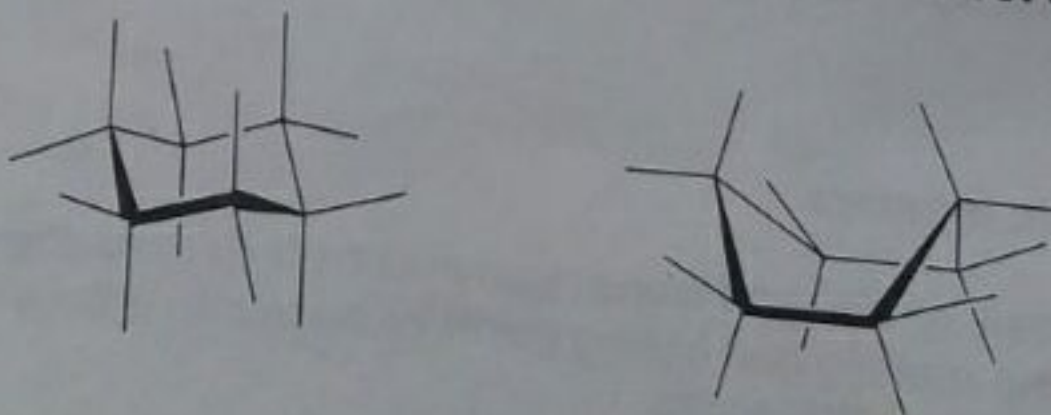


2. Conformational isomers

2.1 Build model for n-butane. Observe the anti, gauche and eclipsed conformations of this molecule. Draw the Newman projection of each form. Why is the eclipsed conformation the most unstable conformer?



2.2 Construct the chair conformation of 1-chlorocyclohexane molecule. Show which bonds are equatorial and which are axial. There are two chair conformers for this molecule. Which form is more stable? Why? In reality this transformation occurs at a very fast rate. Explain why this transformation is very slow when there is a bulky substituent like tert-butyl group. Then make the boat conformer. Why is this most unstable?



Experiment 9: The reaction flask should not be allowed to cool to room temperature because the soap would then solidify and it would consequently be difficult to get it out of the round bottom flask. It may be worthwhile to discuss use of other starting materials besides beef tallow for the preparation of soap.

Experiment 10: Dilute solutions ca. 1% of the four indicators should be prepared in ethanol and placed in small vials. The structures of the dyes should be written on the black board so that the students will reproduce them in their report. The demonstration of TLC and CC is an important component of this experiment. Even used plates obtained from research laboratories would do for the demonstrations. Separation of coloured substances could then be demonstrated using the available TLC plate. If a UV lamp is available in the laboratory, colourless substances could be separated and then viewed by the UV lamp. For the demonstration of column chromatography a small column packed with silica gel is needed. Even a Pasteur pipette filled with silica can be used for a quick demonstration.

Experiment 12: The phenylhydrazine or 2,4-DNPH reagent is prepared by taking 10 mL of phenylhydrazine and adding to it 30 mL of glacial acetic acid and 100 mL of water.

2,4-DNPH reagent is prepared by dissolving 0.5 g DNPH in 80 mL of conc HCl and 100 mL of distilled water. Apply gentle heat on water bath, cool, add 120 mL of water and filter.

The Molisch reagent is prepared by dissolving 15 g of alpha-naphtol in 85 mL of ethanol and then adding ethanol to 100 mL.

Fehling's solution: Prepare **solution 1** from 34.6 g of copper sulphate pentahydrate placed in 500 mL volumetric flask to which a few drops of sulfuric acid are added and then more water to the 500 mL mark. **Solution 2:** 173 g of sodium potassium tartarate and 60 gm of sodium hydroxide dissolved in water and diluted to 500 mL of water. The reagent, Fehling's solution is made just prior to use by mixing equal volumes of solution 1 and 2.

Benedict's solution: Dissolve 86.5 g of sodium citrate and 50 g of anhydrous sodium carbonate in about 350 mL of water, filter if necessary. To this add a solution made up of 8.6 g of copper sulphate pentahydrate and 50 mL of water with constant stirring. Dilute to 500 mL and a clear solution should result, if not filter.

Experiment 13: Acetanilide, aniline, benzaldehyde, cinnamic acid, glycine, glucose, fructose, lactose, maleic acid, maleic anhydride, stearic acid. The benzaldehyde could be supplied as a 30% solution in ethanol. Samples should be issued to students by placing them in small vials (penicillin vials are by far the cheapest and easy to get hold of). Unused samples should be kept for later use.